Financial Data Science

A Unified Modeling Framework

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A self-contained master guide organized around one modeling system:
Model → Fit → Uncertainty → Diagnose → Select → Repair → Predict.

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Contents

1 Part I — What Does It Mean to Model Financial Data? 7
1.1 What model fitting is . . . 7
1.2 Model class, bias, variance, and overfitting . . . 7
1.2.1 Double descent . . . 8
1.3 Parameters, statistics, estimators, and estimates . . . 8
1.4 Bias, variance, standard error, and MSE . . . 8
1.5 Estimator comparison: mean example . . . 8
1.6 Estimating a variance: why divide by $n-1$? . . . 9
1.7 Shrinkage for variance estimation . . . 9
1.8 Reporting uncertainty and confidence intervals . . . 9

2 Part II — Building the Model: Mean, Scale, and Innovation Structure 10
2.1 The three-component architecture . . . 10
2.2 A. Mean structure: different columns, one design-matrix idea . . . 10
2.3 Mean models . . . 10
2.4 Model and assumptions . . . 10
2.5 Multiple linear regression . . . 11
2.6 Factor-model example: Fama–French . . . 11
2.7 Discrete does not always mean categorical . . . 11
2.8 Dummy/indicator coding . . . 11
2.9 Two categorical predictors . . . 11
2.10 Interactions . . . 12
2.11 Yield curves and the Nelson–Siegel model . . . 12
2.12 Nelson–Siegel–Svensson and model flexibility . . . 12
2.13 Option calibration and implied volatility . . . 12
2.14 B. Scale / variance structure . . . 13
2.15 C. Innovation / error-distribution structure . . . 13
2.16 Financial example: log returns and a normal model . . . 13
2.17 Generalized normal distribution as a more flexible tail model . . . 14
2.18 Alternatives to the lognormal pricing model . . . 14
2.19 Generalized Error Distribution (GED) . . . 14
2.20 Nonstandard t distribution . . . 14
2.21 Skewed t distribution . . . 15

3 Part III — Fitting the Model 15
3.1 A. Least-squares family and the general linear model . . . 15
3.2 Least squares as a fitting rule . . . 15
3.3 L_2 versus L_1 fitting . . . 15
3.4 Weighted least squares . . . 16
3.5 Design matrix representation . . . 16
3.6 Hat matrix . . . 16
3.7 Fitted values and residuals . . . 17
3.8 Geometry of least squares . . . 17

3.9 OLS parameter estimates . . . 17
3.10 Covariance and correlation . . . 17
3.11 Regressing Y on X is not the inverse of regressing X on Y . . . 17
3.12 Estimating the error variance . . . 18
3.13 B. Maximum likelihood as the general probability-model fitting rule . . . 18
3.14 Likelihood: reverse the viewpoint . . . 18
3.15 IID exponential example . . . 18
3.16 Log-likelihood . . . 19
3.17 Normal errors connect MLE to least squares . . . 19
3.18 Numerical likelihood maximization with `scipy.optimize.minimize` . . . 19
3.19 Two-parameter Gamma example . . . 20
3.20 Regression with heavy-tailed t errors . . . 20
3.21 GARCH(1,1): likelihood with conditional variances . . . 20

4 Part IV — Quantifying Estimation Uncertainty 21
4.1 Chi-squared distribution . . . 21
4.2 t distribution . . . 21
4.3 Classic normal-sample results . . . 22
4.4 Distribution and covariance results . . . 22
4.5 Degrees of freedom in multiple regression . . . 22
4.6 Inference for the simple-regression slope . . . 22
4.7 Curvature and uncertainty . . . 23
4.8 Approximate normality of a one-parameter MLE . . . 23
4.9 Exponential example: variance shrinks like $1/n$ . . . 23
4.10 Regularity conditions and when asymptotic normality can fail . . . 24
4.11 Counterexample: Uniform$(0, \theta)$ . . . 24
4.12 Multiple parameters: Hessian and covariance matrix . . . 24
4.13 Numerical Hessian and Gamma example . . . 25
4.14 Fisher information, asymptotic bias, and Cramér–Rao bound . . . 25
4.15 Estimating functions of parameters and MLE invariance . . . 25
4.16 Reparameterization can improve numerical optimization . . . 26
4.17 Delta Method: one-dimensional case . . . 26
4.18 Delta Method: multiple parameters . . . 26

5 Part V — Evaluating the Fitted Model 27
5.1 Three evaluation targets . . . 27
5.2 Coefficient of determination . . . 27
5.3 Anscombe's quartet: numerical summaries are not enough . . . 27
5.4 Duplicating every observation: pseudo-replication . . . 27
5.5 What can happen when a predictor is added? . . . 28

6 Part VI — Selecting Model Complexity 28
6.1 Why predictor selection is needed . . . 28
6.2 Akaike Information Criterion (AIC) . . . 28
6.3 Polynomial forward-rate example and scaling . . . 29
6.4 Exhaustive search . . . 29

6.5 Stepwise regression . . . 29
6.6 Adjusted R^2, BIC, and Mallows' C_p . . . 29
6.7 Leave-one-out cross-validation and PRESS . . . 30
6.8 Trace of the hat matrix as model complexity . . . 30
6.9 AIC versus PRESS . . . 30
6.10 LASSO . . . 30
6.11 Why likelihood alone is not enough . . . 31
6.12 Fitting the four return models by maximum likelihood . . . 31
6.13 Nested models and overfitting . . . 31
6.14 General AIC for probability models . . . 31
6.15 Selecting a GARCH model by AIC . . . 32

7 Part VII — Diagnosing Model Failure 32
7.1 Residual diagnostics . . . 32
7.2 Normal Q–Q plots . . . 33
7.3 Multicollinearity . . . 33
7.3.1 Variance inflation factor (VIF) . . . 33
7.4 Influential observations and Cook's distance . . . 33
7.4.1 How to handle an influential point . . . 34

8 Part VIII — Repairing the Model 34
8.1 A. Repairing mean-structure misspecification . . . 34
8.2 B. Outliers and heavy tails . . . 34
8.3 Robust regression and Huber loss . . . 34
8.4 C. Heteroskedasticity: three distinct repair strategies . . . 35
8.5 Heteroskedasticity . . . 35
8.5.1 1. Transform the response . . . 35
8.5.2 2. Weighted least squares and feasible GLS . . . 35
8.5.3 3. Heteroskedasticity-consistent (sandwich) standard errors . . . 35
8.6 D. Multicollinearity responses . . . 36
8.7 E. Power transformations and extreme scales . . . 36
8.8 Motivation . . . 36
8.9 Box–Cox transformation . . . 36
8.10 Box–Cox for regression is not the same as normalizing Y alone . . . 36
8.11 Yeo–Johnson transformation . . . 36
8.12 MSFT spread–volume example: full diagnostic workflow . . . 37

9 Part IX — Prediction 37
9.1 Point prediction . . . 37
9.2 Prediction error variance . . . 37
9.3 Prediction interval . . . 37
9.4 Why normality can still matter for a large prediction sample . . . 38

10 Part X — Financial Quantities Derived from the Fitted Model 38
10.1 Risk-management application: VaR and CVaR . . . 38

11 Implementation and API Cues **39**

12 Practical Q&A and Problem-Solving Guide **40**
12.1 Question-type map . . . 40
12.2 Interpreting Python output and figures . . . 40
12.3 Deriving a likelihood function . . . 41
12.4 Deriving MSE and explaining bias–variance tradeoff . . . 42
12.5 Situation questions: diagnose first, then choose the remedy . . . 43
12.6 Explaining concepts clearly . . . 43
12.7 Constructing and interpreting confidence intervals . . . 44
12.8 Core derivations and distinctions to know fluently . . . 44
12.9 Practical mastery checklist . . . 44

13 Financial Data Science: Complete Modeling Map **45**
13.1 Full logical workflow . . . 45
13.2 Model architecture menu . . . 46
13.3 Fitting-method menu . . . 46
13.4 Evaluation menu . . . 46
13.5 Problem → symptom → consequence → remedy . . . 47
13.6 Prediction and inference map . . . 48
13.7 Complete concept checklist . . . 48
13.8 Complete formula checklist . . . 48

Central architecture. Financial data science approximates an unknown data-generating process by separating systematic behavior, uncertainty size, and standardized randomness:

$$Y_t = m_\theta(X_t, \mathcal{F}_{t-1}) + s_\eta(X_t, \mathcal{F}_{t-1})\epsilon_t, \qquad \epsilon_t \sim G_\phi.$$

The mean structure m_θ describes the conditional center; the scale structure s_η describes how large uncertainty is; and G_ϕ describes the shape of standardized innovations. Many regression problems begin with only the mean model. A complete probability model becomes essential for likelihood, tail modeling, and conditional-volatility models.

Model → Fit → Uncertainty → Diagnose → Select → Repair → Predict

This is an iterative loop rather than a one-way pipeline: prediction and diagnostics can reveal misspecification, which sends us back to revise the model, refit it, and reassess uncertainty and generalization.

1 Part I — What Does It Mean to Model Financial Data?

Before choosing a regression, distribution, or financial time-series model, separate two decisions: the *model class* and the *fitting rule*. The first controls what patterns the model can represent; the second determines how data choose a member of that class. This distinction leads directly to bias, variance, overfitting, estimator quality, and the need to report uncertainty rather than a point estimate alone.

1.1 What model fitting is

A core step in data analysis is to replace a complicated real data-generating process with a mathematically and computationally simpler model. The central attitude is: *all models are wrong, some are useful.* We do not need to believe that the data were literally generated by the chosen probability model; we need the model to approximate the relevant behavior well enough for the task.

Fitting / estimation / training / learning all refer to selecting, from a chosen class of candidate models, the member that best fits a training data set.

Two choices are always present:

1. the **space/class of candidate models**;
2. the **fitting rule** used to choose one member of that class.

These choices drive bias, variance, overfitting, and predictive performance.

1.2 Model class, bias, variance, and overfitting

Imagine the true data-generating process as one point in the space of all possible probability models, and the chosen model class as a smaller region. Repeating the fitting procedure on many data sets would produce a cloud of fitted models.

Bias is systematic displacement of this cloud from the truth: lack of accuracy. **Variance** is the spread of the cloud: lack of precision.

Typical cases:

- **Low bias, high variance:** the model class is flexible enough to contain/approximate the truth, but the fit is unstable across samples.
- **High bias, low variance:** the fit is stable but the class is too restrictive or misspecified.
- **Low bias, low variance:** ideal; often requires a well-chosen class and enough data.
- A lucky restrictive model can have both low bias and low variance, but this is not a general guarantee.

Increasing flexibility commonly reduces bias but increases variance because more parameters must be estimated. Flexible models also increase the risk of **overfitting**: fitting the training sample well but generalizing poorly.

More data typically reduce variance for a fixed flexible class. Large bias caused by an incorrect model class may persist even with a very large sample.

Classical bias–variance tradeoff. More flexibility ⇒ usually lower bias but higher estimator variance. More data ⇒ usually lower variance. A larger sample cannot automatically repair structural model bias.

1.2.1 Double descent

The classical story predicts a U-shaped test-error curve as model complexity grows. Modern highly over-parameterized models can instead show **double descent**: error first falls, then rises near the interpolation threshold, and may fall again after the model has enough capacity to interpolate the training set. In such regimes, adding parameters can again reduce both variance and test error.

1.3 Parameters, statistics, estimators, and estimates

A **parameter** is an unknown constant indexing a probability model. A **statistic** is a function of the observed random variables. An **estimator** is a statistic designed to estimate an unknown parameter and is itself a random variable. An **estimate** is the numerical value obtained after observing one sample.

The hat notation $\hat{\theta}$ denotes an estimator of θ.

1.4 Bias, variance, standard error, and MSE

For an estimator $\hat{\theta}$ of parameter θ,

$$\text{Bias}(\hat{\theta}) = \mathbb{E}(\hat{\theta}) - \theta.$$

It is **unbiased** if $\mathbb{E}(\hat{\theta}) = \theta$.

The estimator variance measures precision:

$$\text{Var}(\hat{\theta}).$$

The **standard error** is

$$\text{SE}(\hat{\theta}) = \sqrt{\text{Var}(\hat{\theta})}.$$

The mean squared error is

$$\text{MSE}(\hat{\theta}) = \mathbb{E}[(\hat{\theta} - \theta)^2] = \text{Var}(\hat{\theta}) + \text{Bias}(\hat{\theta})^2.$$

Thus MSE makes the bias–variance tradeoff explicit.

Estimator versus estimate. Unbiasedness and standard error are properties of the *procedure/estimator*, not of one realized numerical estimate. Do not say "10.5 is unbiased." Say "the estimator used to produce 10.5 is unbiased."

1.5 Estimator comparison: mean example

Suppose $X_1, \ldots, X_n$ are iid with mean μ and variance σ^2. Compare

$$\hat{\mu}_1 = \bar{X}, \qquad \hat{\mu}_2 = \frac{1}{10}\sum_{i=1}^{10} X_i, \qquad \hat{\mu}_3 = 0.$$

Then

$$\mathbb{E}(\bar{X}) = \mu, \qquad \text{Var}(\bar{X}) = \frac{\sigma^2}{n}, \qquad \text{SE}(\bar{X}) = \frac{\sigma}{\sqrt{n}}, \qquad \text{MSE}(\bar{X}) = \frac{\sigma^2}{n}.$$

The average of only ten observations is still unbiased, but has variance $\sigma^2/10$. The constant estimator 0 has zero variance but bias $-\mu$, so its MSE is μ^2. If μ is known a priori to be very near zero, this biased estimator can have lower MSE than a noisy unbiased estimator: a basic example of **shrinkage**.

Normality was not required for these mean/variance calculations; independence and finite moments were the key ingredients.

1.6 Estimating a variance: why divide by $n-1$?

The usual sample variance is

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2.$$

If X_i have common mean μ, common variance σ^2, and are uncorrelated, then s^2 is unbiased for σ^2.
Why not divide by n? Because $\bar{X}$ is chosen from the same data to minimize

$$\sum_{i=1}^{n}(X_i - c)^2.$$

Therefore

$$\sum_{i=1}^{n}(X_i - \bar{X})^2 \le \sum_{i=1}^{n}(X_i - \mu)^2,$$

so division by n systematically underestimates the variance. The factor $n/(n-1)$ corrects this downward bias.
For normal data, the maximum-likelihood estimator is instead

$$\hat{\sigma}^2_{\text{MLE}} = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2,$$

which is biased but arises from likelihood maximization.

Python cue. `np.var(x, ddof=1)` uses $n-1$. `scipy.stats.norm.fit` uses maximum likelihood, so its fitted variance corresponds to the n denominator.

1.7 Shrinkage for variance estimation

If $X_i \overset{iid}{\sim} N(\mu, \sigma^2)$ and we estimate σ^2 using

$$\hat{\sigma}^2_k = k\sum_{i=1}^{n}(X_i - \bar{X})^2,$$

then:

- $k = 1/(n-1)$ gives the unbiased estimator;
- $k = 1/n$ gives the MLE;
- the value minimizing MSE is $k = 1/(n+1)$.

The MSE-optimal estimator is intentionally shrunk downward relative to the unbiased estimator. A key point is that shrinkage becomes especially important in higher-dimensional problems.

1.8 Reporting uncertainty and confidence intervals

Always report not only an estimate but also its uncertainty, usually by a standard error or confidence interval.
A $100(1-\alpha)\%$ confidence interval (L, U) satisfies

$$\Pr(L \le \theta \le U) = 1 - \alpha$$

under repeated sampling. The endpoints L, U are random; the parameter θ is fixed.
If $\hat{\theta}$ is unbiased and approximately normal,

$$\frac{\hat{\theta} - \theta}{\text{SE}(\hat{\theta})} \approx N(0,1),$$

so an approximate 95% interval is

$$\hat{\theta} \pm 1.96\ \text{SE}(\hat{\theta}) \approx \hat{\theta} \pm 2\ \text{SE}(\hat{\theta}).$$

For a large sample mean with known/estimated σ,

$$\bar{X} \pm 2\frac{\sigma}{\sqrt{n}}$$

is the corresponding approximate 95% interval.

A frequentist 95% confidence interval does *not* mean "there is 95% probability that this fixed θ lies in the already computed interval." The 95% refers to the long-run coverage of the random interval construction.

2 Part II — Building the Model: Mean, Scale, and Innovation Structure

A useful stochastic model answers three different questions: what is the expected response, how large is the uncertainty, and what shape does standardized randomness have? Keeping these components separate prevents common confusions such as treating a mean model as a full probability model or treating volatility dynamics as the same thing as a heavy-tailed innovation distribution.

2.1 The three-component architecture

Write

$$Y_t = m_\theta(X_t, \mathcal{F}_{t-1}) + s_\eta(X_t, \mathcal{F}_{t-1})\epsilon_t, \qquad \epsilon_t \sim G_\phi.$$

The first component is the conditional mean; the second is a scale or volatility model; the third is an innovation distribution. A simple homoskedastic regression uses a linear m_θ, constant $s_\eta = \sigma$, and may or may not additionally assume Normal G_ϕ. A GARCH model keeps these roles separate but allows s_η to evolve over time.

2.2 A. Mean structure: different columns, one design-matrix idea

Constant, linear, factor, polynomial, categorical, and interaction models are all choices for the systematic mean. In linear regression, polynomial powers, indicator variables, and interaction products are simply different columns of the design matrix X; they do not require a new fitting theory.

2.3 Mean models

A **mean model** specifies the conditional expectation without necessarily specifying the full probability distribution:

$$\mathbb{E}(Y_i) = m(\theta, x_i),$$

where x_i are known covariates/features and θ is an unknown parameter vector. Equivalently,

$$Y_i = m(\theta, x_i) + \varepsilon_i,$$

where ε_i is a mean-zero error term. A mean model describes the center of the response, not the full scatter around that center.

2.4 Model and assumptions

The simple linear regression model is

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i, \qquad i = 1, \ldots, n.$$

The basic assumptions here are

$$\mathbb{E}(\varepsilon_i) = 0, \qquad \mathrm{Var}(\varepsilon_i) = \sigma^2, \qquad \mathrm{Cov}(\varepsilon_i, \varepsilon_j) = 0 \quad (i \neq j).$$

The second is **homoskedasticity**; the third says errors are uncorrelated. Normal errors are a stronger optional assumption used for exact finite-sample inference.

In finance, the slope in a regression of a stock's excess return on the market excess return is the stock's market **beta**.

2.5 Multiple linear regression

The multiple regression model is

$$Y_i = \beta_0 + \sum_{j=1}^{p} \beta_j X_{ij} + \varepsilon_i,$$

with mean-zero, uncorrelated errors of common variance σ^2. Normality is optional for OLS estimation but enables exact finite-sample inference.

2.6 Factor-model example: Fama–French

A factor model is a multiple linear mean model for returns. The Fama–French three-factor form adds to the market factor:

- **SMB (Small Minus Big):** small-cap minus large-cap portfolio return;
- **HML (High Minus Low):** high book-to-market minus low book-to-market portfolio return.

A stock excess-return mean model can be written as

$$\mathbb{E}(R_i^{\text{excess}}) = \alpha + \beta_M \,\text{MktRF}_i + \beta_S \,\text{SMB}_i + \beta_H \,\text{HML}_i.$$

It is a *mean model*: the equation specifies the conditional mean, not the complete error distribution.

2.7 Discrete does not always mean categorical

A count variable such as number of credit cards or number of dependents is discrete but can often be used as an ordinary quantitative predictor. If it is on a ratio scale, the regression interpretation remains: a one-unit increase changes the conditional mean linearly according to its slope coefficient.

An **ordinal** variable (for example, education coded 0,1,2,3) should not automatically be treated as quantitative. Doing so assumes:

- equal mean differences between adjacent levels;
- moving 0 to 2 has exactly twice the effect of moving 0 to 1.

These assumptions are often implausible.

2.8 Dummy/indicator coding

A categorical variable with k levels adds $k-1$ coefficient parameters when an intercept is present. Choose one level as the **baseline/reference**. For education levels $0, 1, 2, 3$ and quantitative predictor X,

$$\mathbb{E}(Y) = \beta_0 + \beta_1 I\{\text{edu} = 1\} + \beta_2 I\{\text{edu} = 2\} + \beta_3 I\{\text{edu} = 3\} + \beta_4 X.$$

The three education coefficients are shifts relative to level 0. With no interactions, each education group has the same slope in X but a different intercept.

Treating every discrete variable as categorical is not automatically better: it adds many parameters, raises overfitting risk, and requires enough data at every category level.

Python cues. Pandas categorical dtype can be used with formula-based regression. `pd.get_dummies(..., drop_first=True)` creates $k-1$ indicator columns and avoids a redundant dummy column when an intercept is present.

2.9 Two categorical predictors

If categorical X_1 has k_1 levels and X_2 has k_2 levels, an additive model with an intercept has

$$1 + (k_1 - 1) + (k_2 - 1)$$

intercept/main-effect parameters before adding interactions. Each coefficient is interpreted relative to the baseline levels, holding the other variable fixed.

2.10 Interactions

An interaction is formed by multiplying predictors. Without interactions, the model is **additive**: the effect of one predictor is assumed not to depend on the value of another.

For two quantitative predictors,

$$\mathbb{E}(Y) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 X_2.$$

Holding X_1 fixed, the slope with respect to X_2 is

$$\beta_2 + \beta_3 X_1.$$

Thus the effect of X_2 varies with X_1.

With a quantitative predictor and a categorical predictor, interaction terms allow category-specific slopes rather than only parallel shifted lines. Main effects are then conditional: the main effect of the quantitative variable corresponds to the baseline category, and categorical main effects correspond to differences at the quantitative value zero (or at whatever centering point is used).

A single combined interaction model can be preferable to fitting entirely separate regressions by category because it pools information, especially for estimating a common error variance when that assumption is appropriate.

For two categorical predictors represented by indicator columns, products of the indicators create categorical-by-categorical interactions. Such an interaction compares how one category contrast changes across the levels of the other variable—a **difference of differences**. More generally, once an interaction is present, main effects are conditional on the other predictor being at its reference value or numerical zero. Centering a continuous predictor moves that zero point to a meaningful value (for example its sample mean), which changes the interpretation of the main effects but not the fitted interaction surface.

2.11 Yield curves and the Nelson–Siegel model

A yield curve plots yields against maturity. The Nelson–Siegel mean model is

$$y(T) = \beta_0 + \beta_1 \left(\frac{1 - e^{-T/\tau}}{T/\tau} \right) + \beta_2 \left(\frac{1 - e^{-T/\tau}}{T/\tau} - e^{-T/\tau} \right).$$

Parameters $(\beta_0, \beta_1, \beta_2, \tau)$ control the level and shape of the curve. This is a mean model: it specifies the curve, not the probability distribution of observations around the curve.

A representative application fits the model to Treasury par-yield data by minimizing squared residuals, either directly with a numerical optimizer or with a package calibration routine.

Python cues. A generic nonlinear least-squares objective can be passed to `scipy.optimize.minimize`. The Nelson–Siegel package also supplies calibration utilities such as an OLS calibration routine. Good initialization matters in nonlinear optimization.

2.12 Nelson–Siegel–Svensson and model flexibility

The Nelson–Siegel–Svensson extension adds another curvature term, typically with an additional coefficient and decay parameter:

$$y(T) = \text{Nelson–Siegel terms} + \beta_3 \left(\frac{1 - e^{-T/\tau_2}}{T/\tau_2} - e^{-T/\tau_2} \right).$$

The more flexible model can reduce RSS, but a lower training RSS does not by itself prove better predictive performance. If a simpler model misses systematic shape in the data, extra flexibility can help; if not, the larger model may merely overfit.

2.13 Option calibration and implied volatility

The Black–Scholes model uses five principal inputs:

1. spot price S;
2. strike K;

3. time to expiration T;
4. risk-free rate r;
5. volatility σ.

Volatility is not directly observed. Given an observed option market price, the **implied volatility** is the value of σ that makes the Black–Scholes price equal to the market price.

Under constant-volatility Black–Scholes, options with the same underlying and expiration would share the same implied volatility. In real data, implied volatility varies with strike, producing a **volatility smile/skew**. More flexible option models (for example, stochastic-volatility models such as Heston) can be calibrated to this structure.

Why calibrate the smile rather than raw option prices? Raw option prices can span a very large range across strikes. Least squares on raw prices can be dominated by the large-price options. Implied volatility puts the observations onto a more comparable scale and makes the shape of the volatility surface directly visible.

Weighted least squares can again be useful: heavily traded options can be given more weight because their market prices may be more reliable.

2.14 B. Scale / variance structure

Under homoskedasticity,

$$\mathrm{Var}(\epsilon_i \mid X) = \sigma^2,$$

whereas under heteroskedasticity,

$$\mathrm{Var}(\epsilon_i \mid X) = \sigma_i^2.$$

This distinction can affect efficiency and, more importantly, the covariance formula used for standard errors, hypothesis tests, and confidence intervals. In financial time series the analogous idea is time-varying scale,

$$X_t = \mu + \sigma_t \epsilon_t,$$

which motivates conditional-volatility models such as GARCH. The detailed diagnostics and remedies for nonconstant variance appear after the fitting and uncertainty machinery is established.

2.15 C. Innovation / error-distribution structure

Once the center and scale have been separated, the remaining question is the shape of standardized randomness. Normal, GED, t, and skewed-t distributions represent progressively different tail and asymmetry behavior. This choice matters little for writing a mean function but becomes central for likelihood, exact finite-sample inference, prediction tails, and financial risk.

2.16 Financial example: log returns and a normal model

For an asset price P_t, the log return is

$$r_t = \log\left(\frac{P_t}{P_{t-1}}\right).$$

Using $\log(1+x) \approx x$ for small x,

$$\log\left(\frac{P_t}{P_{t-1}}\right) = \log\left(1 + \frac{P_t - P_{t-1}}{P_{t-1}}\right) \approx \frac{P_t - P_{t-1}}{P_{t-1}},$$

so log returns are close to simple returns when returns are small.

A representative application fits a normal distribution to daily log returns and compare the empirical histogram with the fitted density. A normal model can be mathematically convenient, but the return tails are visibly imperfect: on a log-density scale, real return data can display substantially more tail mass than a fitted normal distribution.

Python cues. `yfinance` can download market data, but it has limitations (including survivorship bias). By default, adjusted prices account for splits and dividends. In `scipy.stats`, `norm.fit` estimates the normal parameters by maximum likelihood; distribution objects also provide PDF/CDF/SF methods.

Model assumption versus reality. A useful model need not be literally true. But if a tail probability, confidence interval, or risk estimate depends strongly on a normal-tail assumption, a poor tail fit matters materially.

2.17 Generalized normal distribution as a more flexible tail model

A flexible symmetric alternative to the normal is the generalized normal density

$$f(x) = \frac{\beta}{2\alpha\,\Gamma(1/\beta)} \exp\left[-\left(\frac{|x-\mu|}{\alpha}\right)^{\beta}\right], \qquad -\infty < x < \infty,$$

where $\alpha > 0$ is a scale parameter, $\beta > 0$ controls tail shape, and μ is the center. Its variance is

$$\mathrm{Var}(X) = \alpha^2 \frac{\Gamma(3/\beta)}{\Gamma(1/\beta)}.$$

For this parameterization, $\beta = 2$ gives the normal shape; $\beta > 2$ produces lighter tails and $\beta < 2$ produces heavier tails. This illustrates the tradeoff: a larger model class can fit observed tails better but requires estimating more parameters.

2.18 Alternatives to the lognormal pricing model

Let S_t be an equity closing price. The lognormal pricing model writes

$$S_t = S_{t-1} X_t, \qquad X_t \sim \text{Lognormal}(\mu, \sigma^2),$$

with independent X_t. Hence

$$\log\left(\frac{S_t}{S_{t-1}}\right) = \log X_t \sim N(\mu, \sigma^2).$$

the examples examine Procter & Gamble (PG) daily log returns from 2015–2025. A normal Q–Q plot bends strongly in both tails, and a log-density comparison makes the discrepancy especially visible: empirical log returns have much heavier tails than a fitted normal distribution.

Why use a log density scale? Tail densities can be tiny on the ordinary scale. A logarithmic vertical axis magnifies relative tail differences, making underestimation of extreme-event probability visually obvious.

2.19 Generalized Error Distribution (GED)

The **GED**, also called the generalized normal distribution, is the same family introduced earlier as a flexible symmetric alternative to the normal:

$$f_X(x) = \frac{\beta}{2\alpha\Gamma(1/\beta)} \exp\left[-\left(\frac{|x-\mu|}{\alpha}\right)^{\beta}\right], \qquad \alpha > 0,\ \beta > 0.$$

Here μ is the center, α is scale, and β controls shape/tails.

- $\beta = 2$: $N(\mu, \alpha^2/2)$;
- $\beta = 1$: double-exponential/Laplace distribution;
- the family is symmetric around μ;
- in SciPy it is `scipy.stats.gennorm`.

Changing β changes the tail/peak shape; this added shape parameter is exactly the flexibility that can improve fit but also raises model complexity.

2.20 Nonstandard t distribution

If T has a standard t distribution with ν degrees of freedom, define

$$X = \mu + \sigma T, \qquad \sigma > 0.$$

This keeps the heavy-tailed t shape while allowing arbitrary location μ and scale σ. In Python this is represented by `scipy.stats.t`. Since

$$\mathrm{Var}(T) = \frac{\nu}{\nu - 2} \quad (\nu > 2),$$

we have

$$\mathrm{Var}(X) = \sigma^2 \frac{\nu}{\nu - 2}.$$

Thus the parameter called "scale" is not the standard deviation; the standard deviation is larger by the factor $\sqrt{\nu/(\nu-2)}$.

2.21 Skewed t distribution

Jones and Faddy's skewed t generalizes the t distribution to allow asymmetry through positive parameters a and b:

- $a < b$: negatively skewed;
- $a > b$: positively skewed;
- $a = b$: reduces to the nonstandard t with $2a$ degrees of freedom.

One property shown here is

$$\mathbb{E}(T) = \mu + \frac{(a-b)\sqrt{a+b}\,\Gamma(a-\frac{1}{2})\Gamma(b-\frac{1}{2})}{2\Gamma(a)\Gamma(b)}.$$

SciPy provides `scipy.stats.jf_skew_t`. This model adds flexibility for asymmetric left and right tails.

3 Part III — Fitting the Model

After specifying the model class, the estimation problem is to choose parameter values. Least squares is a fitting rule for a mean model; maximum likelihood is the general fitting rule once a full probability model is specified. The linear-regression matrix framework connects the two.

3.1 A. Least-squares family and the general linear model

3.2 Least squares as a fitting rule

For training pairs (x_i, y_i), least squares chooses θ to minimize

$$\mathrm{RSS}(\theta) = \sum_{i=1}^{n} \big(y_i - m(\theta, x_i)\big)^2.$$

The motivation is geometric: choose the model curve whose fitted values are, in squared distance, closest to the observations.

If the mean model is simply $m(\mu, x) = \mu$, least squares minimizes

$$\sum_{i=1}^{n} (y_i - \mu)^2,$$

and the minimizer is

$$\hat{\mu}_{LS} = \bar{Y}.$$

Least squares is therefore much broader than ordinary linear regression.

3.3 L_2 versus L_1 fitting

Least squares uses an L_2 loss:

$$\sum_{i=1}^{n} \big(y_i - (\beta_0 + \beta_1 x_i)\big)^2.$$

An L_1 alternative minimizes

$$\sum_{i=1}^{n} |y_i - (\beta_0 + \beta_1 x_i)| .$$

L_2 is smooth and yields simple closed-form expressions in linear regression. L_1 is less sensitive to heavy-tailed errors and extreme observations but generally requires numerical optimization.

3.4 Weighted least squares

Ordinary least squares (OLS) minimizes

$$\sum_i (y_i - m_i)^2.$$

Weighted least squares (WLS) minimizes

$$\mathrm{WRSS}(\theta) = \sum_{i=1}^{n} w_i \big(y_i - m(\theta, x_i)\big)^2.$$

If observation i has error variance σ_i^2, the natural/optimal weight is

$$w_i = \frac{1}{\sigma_i^2}.$$

Observations with greater uncertainty receive less weight. In the yield-curve example, less liquid maturities might reasonably receive lower weight than highly liquid maturities.

3.5 Design matrix representation

Simple and multiple linear regression can be written as

$$\mathbf{Y} = X\boldsymbol{\beta} + \boldsymbol{\varepsilon},$$

where X is the design matrix. With an intercept and p predictors, X is $n \times (p+1)$ and the first column is all ones.
The OLS estimator is

$$\hat{\boldsymbol{\beta}} = (X^T X)^{-1} X^T \mathbf{Y},$$

provided $X^T X$ is invertible.
Then

$$\hat{\mathbf{Y}} = X\hat{\boldsymbol{\beta}}, \qquad \hat{\boldsymbol{\varepsilon}} = \mathbf{Y} - \hat{\mathbf{Y}}.$$

3.6 Hat matrix

Define

$$H = X(X^T X)^{-1} X^T.$$

Then

$$\hat{\mathbf{Y}} = H\mathbf{Y}, \qquad \hat{\boldsymbol{\varepsilon}} = (I - H)\mathbf{Y}.$$

H is a projection matrix: it is symmetric and idempotent,

$$H^T = H, \qquad H^2 = H.$$

Likewise, $I - H$ projects onto the residual space.
The diagonal entries h_{ii} are **leverages**. They measure how unusual observation i is in predictor space and how strongly its own response can affect its fitted value.

3.7 Fitted values and residuals

After estimating (β_0, β_1),

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i, \qquad \hat{\varepsilon}_i = y_i - \hat{y}_i.$$

Residuals are estimated errors. They are diagnostic objects and obey special algebraic constraints under OLS.

3.8 Geometry of least squares

Think of $\mathbf{y} = (y_1, \ldots, y_n)^T$ as a vector in $\mathbb{R}^n$. With an intercept and one predictor, all possible fitted vectors lie in the two-dimensional subspace spanned by the vectors $\mathbf{1}$ and $\mathbf{x}$. OLS is the **orthogonal projection** of $\mathbf{y}$ onto this subspace.

Therefore the residual vector $\hat{\boldsymbol{\varepsilon}} = \mathbf{y} - \hat{\mathbf{y}}$ is orthogonal to both $\mathbf{1}$ and $\mathbf{x}$:

$$\mathbf{1}^T \hat{\boldsymbol{\varepsilon}} = 0, \qquad \mathbf{x}^T \hat{\boldsymbol{\varepsilon}} = 0.$$

In particular,

$$\sum_i \hat{\varepsilon}_i = 0.$$

This is an algebraic result of OLS with an intercept; it is not the probabilistic assumption $\mathbb{E}(\varepsilon_i) = 0$.

3.9 OLS parameter estimates

Minimizing RSS yields

$$\hat{\beta}_1 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2} = \frac{s_{xy}}{s_x^2} = r_{xy} \frac{s_y}{s_x},$$
$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

If variables are centered and standardized ($\bar{x} = \bar{y} = 0$, $s_x = s_y = 1$), then $\hat{\beta}_0 = 0$ and $\hat{\beta}_1 = r_{xy}$.

Why OLS is used so often:

- the squared-error objective is differentiable and gives closed-form estimators in linear regression;
- under the Gauss–Markov assumptions, OLS estimators are **BLUE**: best linear unbiased estimators;
- if errors are iid normal, OLS coincides with maximum likelihood.

3.10 Covariance and correlation

For random variables X, Y,

$$\mathrm{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}X)(Y - \mathbb{E}Y)], \qquad \mathrm{Corr}(X, Y) = \frac{\mathrm{Cov}(X, Y)}{\sigma_X \sigma_Y}.$$

Sample covariance is

$$s_{xy} = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y}),$$

and sample correlation is

$$r_{xy} = \frac{s_{xy}}{s_x s_y}.$$

Correlation measures **linear association**. A value $|r|$ close to one does not guarantee that a straight-line model is adequate: a strongly curved relation can still have very high correlation.

3.11 Regressing Y on X is not the inverse of regressing X on Y

If Y is regressed on X, the slope is

$$\hat{\beta}_{Y|X} = r_{xy} \frac{s_y}{s_x}.$$

If the roles are reversed,

$$\hat{\beta}_{X|Y} = r_{xy}\frac{s_x}{s_y}.$$

These are not generally reciprocals. OLS minimizes *vertical* residuals in one direction, and horizontal residuals in the reversed direction. The "SD line" with slope s_y/s_x is a different object.

3.12 Estimating the error variance

With two fitted parameters (β_0, β_1), the unbiased estimator is

$$\hat{\sigma}^2 = \frac{1}{n-2}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 = \frac{\text{RSS}}{n-2}.$$

Under iid normal errors, the MLE is RSS/n, which is biased downward. The $n-2$ denominator reflects two fitted regression parameters (two constraints / lost degrees of freedom).

General versus special case. Multiple linear regression $Y = X\beta + \epsilon$ is the central linear framework. Simple linear regression is the one-predictor special case. The familiar scalar slope, intercept, and slope-variance formulas are therefore simplifications of the matrix equations, not separate theories.

3.13 B. Maximum likelihood as the general probability-model fitting rule

Once a full distribution for the data or errors is specified, likelihood evaluates how compatible each parameter value is with the observed sample. This extends fitting beyond squared loss and makes the assumed error distribution part of the estimation rule.

3.14 Likelihood: reverse the viewpoint

A full probability model gives a density or probability law $f(x \mid \theta)$. Usually we fix θ and view this as a function of possible data x. For maximum likelihood, the observed data are fixed and the same expression is viewed as a function of the unknown parameter:

$$L(\theta) = f(x \mid \theta).$$

The **maximum likelihood estimator (MLE)** chooses the parameter value that makes the observed data most plausible under the assumed model:

$$\hat{\theta} = \arg\max_{\theta} L(\theta).$$

For continuous data, an exact point has probability zero. Therefore likelihood should not be interpreted as the probability that X equals the exact observed value; density is large in regions where observations near the realized data are more likely.

Likelihood is not a probability distribution for the parameter. A plot of $L(\theta)$ against θ does not integrate to one and does not assign probabilities to possible parameter values. It compares how well different parameter values explain the fixed observed sample.

3.15 IID exponential example

Suppose $X_1, \ldots, X_n \overset{iid}{\sim} \text{Exponential}(\lambda)$, with

$$f_X(x) = \lambda e^{-\lambda x}, \qquad x > 0,\ \lambda > 0.$$

Independence gives

$$L(\lambda) = \prod_{i=1}^{n} \lambda e^{-\lambda x_i} = \lambda^n \exp\left(-\lambda \sum_{i=1}^{n} x_i\right).$$

For the five observations used here, $0.92, 0.54, 1.20, 0.051, 1.13$, the likelihood is maximized near $\hat{\lambda} = 1.3$.

3.16 Log-likelihood

Because $\log(\cdot)$ is strictly increasing, maximizing $L(\theta) > 0$ is equivalent to maximizing the **log-likelihood**

$$\ell(\theta) = \log L(\theta).$$

For independent data, products become sums:

$$\ell(\theta) = \sum_{i=1}^{n} \log f(x_i \mid \theta).$$

This is computationally preferable because products of many densities can underflow toward zero or overflow, while sums of log-densities stay numerically manageable. The likelihood and log-likelihood have the same maximizer.

For the exponential model,

$$\ell(\lambda) = n \log \lambda - \lambda \sum_{i=1}^{n} x_i, \qquad -\ell(\lambda) = -n \log \lambda + \lambda \sum_{i=1}^{n} x_i.$$

In practice, software usually *minimizes the negative log-likelihood*.

3.17 Normal errors connect MLE to least squares

Suppose

$$Y_i = m(\theta, x_i) + \varepsilon_i, \qquad \varepsilon_i \overset{iid}{\sim} N(0, \sigma^2).$$

Then

$$Y_i \sim N(m(\theta, x_i), \sigma^2)$$

and independence gives

$$L(\theta, \sigma) = \left(\frac{1}{\sqrt{2\pi}\sigma} \right)^n \exp \left[-\frac{1}{2\sigma^2} \sum_{i=1}^{n} \big(y_i - m(\theta, x_i)\big)^2 \right].$$

Thus

$$\ell(\theta, \sigma) = n \log \left(\frac{1}{\sqrt{2\pi}\sigma} \right) - \frac{1}{2\sigma^2} \sum_{i=1}^{n} \big(y_i - m(\theta, x_i)\big)^2.$$

For fixed σ, maximizing the log-likelihood over θ is exactly equivalent to minimizing

$$\sum_{i=1}^{n} \big(y_i - m(\theta, x_i)\big)^2.$$

Therefore **least squares is maximum likelihood when the errors are iid normal with common variance**. This explains one major statistical justification for least squares.

3.18 Numerical likelihood maximization with `scipy.optimize.minimize`

Closed-form MLEs are unavailable for many models. A general procedure is:

1. Write a function returning the **negative log-likelihood**.
2. Choose starting value(s) in the interior of the allowed parameter space.
3. Minimize numerically.
4. Read the optimizer from `res.x` and verify successful convergence.

Core syntax used here.

```
from scipy.optimize import minimize
res = minimize(neg_loglike, start, args=(data,),
               method='Nelder-Mead', options={'disp': True})
hat_theta = res.x
```

Important argument rules:

1. In the function being minimized, the **first argument** must contain the parameter(s) being optimized. If there are several parameters, pass them as an array/vector.
2. Other data/inputs begin with the second function argument and are passed through `args=(...)`.
3. The first argument of `minimize` is the *name* of the objective function, without parentheses.
4. The second argument is the starting value/vector. Choose an interior value, not a boundary value when possible.
5. `method` selects the optimization algorithm; a representative implementation uses Nelder–Mead for the basic examples.
6. `options` controls settings such as iteration limits, tolerance, and displayed output; available options depend on the chosen method.
7. `res.x` contains the parameter value at the numerical minimum.

3.19 Two-parameter Gamma example

The Gamma example uses parameters (α, β) with β as a **rate**, so in SciPy the corresponding scale is $1/\beta$. Its mean is

$$\mathbb{E}(X) = \frac{\alpha}{\beta}.$$

A compact negative log-likelihood implementation is based on

$$-\sum_{i=1}^{n} \log f_\Gamma(x_i; \alpha, \beta),$$

using `scipy.stats.gamma.logpdf(..., a=alpha, scale=1/beta)`. With multiple unknowns, the first optimizer argument is a parameter vector such as `[alpha,beta]`. Simulation here illustrates that estimates improve as the sample size grows.

3.20 Regression with heavy-tailed t errors

the examples replace normal regression errors by a Student-t model:

$$Y_i = \beta_0 + \beta_1 x_i + \sigma\varepsilon_i, \qquad \varepsilon_i \sim t_\nu,$$

where ν is treated as fixed and the unknown parameters are $(\beta_0, \beta_1, \sigma)$. Because scaling by σ changes a density by the Jacobian factor $1/\sigma$,

$$f_{Y_i}(y_i) = \frac{1}{\sigma} f_{t_\nu}\left(\frac{y_i - (\beta_0 + \beta_1 x_i)}{\sigma}\right).$$

Hence, apart from notation, the negative log-likelihood has the structure

$$-\ell(\beta_0, \beta_1, \sigma) = -\sum_{i=1}^{n} \log f_{t_\nu}\left(\frac{y_i - (\beta_0 + \beta_1 x_i)}{\sigma}\right) + n \log \sigma.$$

The implementation uses `scipy.stats.t.logpdf`. A representative implementation uses least-squares coefficient estimates and a residual-scale estimate as convenient starting values. Under heavy-tailed errors, the likelihood model that correctly uses t errors can be more stable than ordinary least squares, because extreme residuals are less incompatible with the assumed error distribution.

3.21 GARCH(1,1): likelihood with conditional variances

Financial return series often show **volatility clustering**: large moves tend to be followed by periods of large moves, and quiet periods by quiet periods. The GARCH(1,1) model here assumes

$$X_t \mid \mathcal{F}_{t-1} \sim N(0, \sigma_t^2), \qquad \sigma_t^2 = \omega + \alpha X_{t-1}^2 + \beta\sigma_{t-1}^2.$$

The unknown parameter vector is

$$\theta = (\omega, \alpha, \beta).$$

This is not iid: the conditional variance changes over time. The joint likelihood is built from conditional densities,

$$f(x_1, \ldots, x_n) = f(x_1) \prod_{t=2}^{n} f(x_t \mid x_1, \ldots, x_{t-1}),$$

and, given the recursively computed σ_t^2, each conditional density is normal.

The initial variance is not specified by the recursion. A convenient initialization uses the stationary/unconditional variance

$$\sigma_1^2 = \frac{\omega}{1-\alpha-\beta},$$

which requires the usual stationary restriction $\alpha + \beta < 1$ together with nonnegative variance parameters.

A compact fitting workflow is:

1. Scale the return series before optimization because raw log returns and especially their squares can be very close to zero.
2. Initialize, for example, with a variance-based guess for ω and moderate positive values for α, β.
3. Recursively compute all σ_t^2.
4. Sum normal `logpdf` values and minimize their negative.
5. Use an optimizer supporting bounds, such as `L-BFGS-B`; the examples constrain ω, α, β to nonnegative ranges and note the stationarity condition $\alpha + \beta < 1$.

Sometimes a numerical lower bound of exactly zero causes difficulty; a tiny positive lower bound such as 10^{-6} can be used instead.

GARCH API cue. Existing packages such as the `arch` package can fit the same model directly. The purpose of the manual likelihood code is to understand the conditional-density construction, parameter constraints, scaling, and optimization, not to replace specialized software.

Unification. Under iid homoskedastic Normal regression errors, maximizing likelihood over the regression coefficients is exactly the same as minimizing RSS. Thus OLS is a special MLE. More broadly, choosing an error distribution chooses a likelihood and therefore a residual-penalty structure: Normal errors imply squared loss; heavy-tailed t errors penalize extremes less aggressively; Huber loss provides a robust compromise without requiring a full probability model.

4 Part IV — Quantifying Estimation Uncertainty

A fitted parameter is only one realization of a random estimator. The next question is how that estimator would vary across repeated samples. Classical χ^2 and t results give exact finite-sample answers in Normal models; matrix regression gives the general OLS covariance; likelihood curvature and Fisher information extend the same idea to general MLEs; the Delta Method then propagates parameter uncertainty into nonlinear functions.

4.1 Chi-squared distribution

If $Z_1, \ldots, Z_k \overset{iid}{\sim} N(0,1)$, then

$$X = \sum_{i=1}^{k} Z_i^2 \sim \chi_k^2.$$

The chi-squared distribution is also a special Gamma distribution. Its mean and variance are

$$\mathbb{E}(X) = k, \qquad \mathrm{Var}(X) = 2k.$$

As the degrees of freedom increase, its density becomes less right-skewed.

4.2 t distribution

If $Z \sim N(0,1)$ and $X \sim \chi_\nu^2$ are independent, then

$$T = \frac{Z}{\sqrt{X/\nu}} \sim t_\nu.$$

The t distribution is symmetric but has heavier tails than the standard normal. As $\nu \to \infty$, t_ν approaches $N(0,1)$. For the cases emphasized here,

$$\mathbb{E}(T) = 0 \quad (\nu \geq 2), \qquad \mathrm{Var}(T) = \frac{\nu}{\nu - 2} \quad (\nu \geq 3).$$

t_1 is the Cauchy distribution, whose tails are so heavy that its mean does not exist and the ordinary law-of-large-numbers intuition for a sample mean fails.

4.3 Classic normal-sample results

If $X_1, \ldots, X_n \overset{iid}{\sim} N(\mu, \sigma^2)$:

1. $\bar{X}$ and s^2 are independent;
2. $(\bar{X} - \mu)/(\sigma/\sqrt{n}) \sim N(0,1)$;
3. $(n-1)s^2/\sigma^2 \sim \chi^2_{n-1}$;
4. $(\bar{X} - \mu)/(s/\sqrt{n}) \sim t_{n-1}$.

The t distribution appears because the unknown σ is replaced by its random estimate s.

4.4 Distribution and covariance results

Under the standard linear model,

$$\mathbb{E}(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta}, \qquad \mathrm{Var}(\hat{\boldsymbol{\beta}}) = \sigma^2 (X^T X)^{-1}.$$

Also,

$$\mathrm{Var}(\hat{\mathbf{Y}}) = \sigma^2 H, \qquad \mathrm{Var}(\hat{\boldsymbol{\varepsilon}}) = \sigma^2 (I - H).$$

If the errors are normal, then

$$\begin{aligned}
&\mathbf{Y} \sim N(X\boldsymbol{\beta}, \sigma^2 I), \\
&\hat{\boldsymbol{\beta}} \sim N\left(\boldsymbol{\beta}, \sigma^2 (X^T X)^{-1}\right), \\
&\hat{\mathbf{Y}} \sim N(X\boldsymbol{\beta}, \sigma^2 H), \\
&\hat{\boldsymbol{\varepsilon}} \sim N(\mathbf{0}, \sigma^2 (I - H)).
\end{aligned}$$

4.5 Degrees of freedom in multiple regression

The residuals satisfy

$$X^T \hat{\boldsymbol{\varepsilon}} = 0.$$

With p predictors plus an intercept, there are $p+1$ fitted coefficients, leaving

$$\boxed{n - p - 1}$$

residual degrees of freedom.

Thus

$$\hat{\sigma}^2 = \frac{\mathrm{RSS}}{n - p - 1},$$

and under normal errors,

$$\frac{\hat{\beta}_j - \beta_j}{\mathrm{SE}(\hat{\beta}_j)} \sim t_{n-p-1}.$$

The usual t-based confidence intervals and hypothesis tests generalize directly from simple to multiple regression using these degrees of freedom.

4.6 Inference for the simple-regression slope

Under the basic regression assumptions,

$$\mathbb{E}(\hat{\beta}_1) = \beta_1.$$

Its variance is

$$\mathrm{Var}(\hat{\beta}_1) = \frac{\sigma^2}{\sum_i (x_i - \bar{x})^2} = \frac{\sigma^2}{(n-1)s_x^2},$$

so

$$\mathrm{SE}(\hat{\beta}_1) \approx \frac{\hat{\sigma}}{s_x\sqrt{n-1}}.$$

If errors are iid normal, then for testing $H_0 : \beta_1 = \beta_{1,0}$,

$$T = \frac{\hat{\beta}_1 - \beta_{1,0}}{\mathrm{SE}(\hat{\beta}_1)} \sim t_{n-2}.$$

Hence a $100(1-\alpha)\%$ confidence interval is

$$\hat{\beta}_1 \pm t_{\alpha/2,n-2}\,\mathrm{SE}(\hat{\beta}_1).$$

For large n, one often uses the approximate normal interval

$$\hat{\beta}_1 \pm z_{\alpha/2}\,\mathrm{SE}(\hat{\beta}_1).$$

For testing $H_0 : \beta_1 = 0$, a 95% interval excluding zero is equivalent to a two-sided p-value below 0.05. "Significant" means the estimated effect is large relative to its standard error; an insignificant result can also occur because the standard error is too large to detect a nonzero effect.

How to reduce the slope SE:

- increase n;
- obtain a larger spread s_x in the predictor;
- reduce the irreducible error scale σ if the data-generating process/design permits.

4.7 Curvature and uncertainty

For a twice-differentiable scalar function $f(t)$ minimized at $\hat{t}$,

$$f'(\hat{t}) = 0, \qquad f''(\hat{t}) > 0$$

for an interior strict local minimum. The magnitude of $f''(\hat{t})$ measures how **sharp** the minimum is: a larger second derivative means the objective rises rapidly away from the minimizer, while a small second derivative means a broad, flat minimum.

For MLE, apply this idea to the **negative log-likelihood**. A sharp minimum means the data pin down the parameter more precisely.

4.8 Approximate normality of a one-parameter MLE

Under suitable regularity conditions, if $\hat{\theta}$ is the MLE of a scalar parameter θ, then for a sufficiently large sample it is approximately normal with mean near the true value and variance approximately equal to the inverse local curvature:

$$\hat{\theta} \approx N\left(\theta, \left[-\frac{d^2\ell(\theta)}{d\theta^2}\bigg|_{\theta=\hat{\theta}}\right]^{-1}\right).$$

Equivalently, the second derivative of the negative log-likelihood is positive and its inverse estimates the MLE variance. A steeper likelihood peak therefore implies a smaller standard error. The approximate normality is an asymptotic result tied to central-limit behavior and is the basis for large-sample confidence intervals and tests.

4.9 Exponential example: variance shrinks like $1/n$

For iid Exponential(λ) data,

$$-\ell(\lambda) = -n\log\lambda + \lambda\sum_{i=1}^{n} x_i.$$

Its second derivative is

$$\frac{d^2[-\ell(\lambda)]}{d\lambda^2} = \frac{n}{\lambda^2}.$$

Therefore

$$\mathrm{Var}(\hat{\lambda}) \approx \frac{\hat{\lambda}^2}{n}.$$

As n grows, the negative log-likelihood becomes sharper and the sampling distribution of the MLE becomes more concentrated.

4.10 Regularity conditions and when asymptotic normality can fail

The phrase "under suitable regularity conditions" includes, in this setting, four main requirements:

1. **Common support:** the support of the observable distribution should not change with θ.
2. **Interior truth:** the true parameter value should not lie on the boundary of the allowed parameter space.
3. **Identifiability:** two distinct parameter values should correspond to two distinct probability models.
4. **Smooth likelihood:** the likelihood/log-likelihood should be sufficiently smooth and differentiable in the parameter.

These assumptions are technical but usually hold in standard smooth models.

4.11 Counterexample: Uniform$(0, \theta)$

If

$$X_1, \ldots, X_n \overset{iid}{\sim} \mathrm{Uniform}(0, \theta),$$

then

$$f_X(x) = \frac{1}{\theta}\mathbf{1}_{\{0 \le x \le \theta\}},$$

and

$$L(\theta) = \theta^{-n}\mathbf{1}_{\{X_{(n)} \le \theta\}}, \qquad X_{(n)} = \max_i X_i.$$

For $\theta < X_{(n)}$ the likelihood is zero; for $\theta \ge X_{(n)}$ it decreases as θ increases. Hence

$$\hat{\theta}_{\mathrm{MLE}} = X_{(n)}.$$

Its CDF and density are

$$\Pr(\hat{\theta} \le t) = \left(\frac{t}{\theta}\right)^n, \qquad f_{\hat{\theta}}(t) = \frac{n}{\theta}\left(\frac{t}{\theta}\right)^{n-1}, \quad 0 \le t \le \theta.$$

This sampling distribution remains one-sided and boundary-concentrated rather than becoming approximately normal in the usual way. The asymptotic-normality formula fails because the support $(0, \theta)$ depends on the unknown parameter. A related correction is the related unbiased "German tank" correction

$$\tilde{\theta} = \frac{n+1}{n} X_{(n)}.$$

4.12 Multiple parameters: Hessian and covariance matrix

For a parameter vector $\theta = (\theta_1, \ldots, \theta_p)^T$, the second derivative generalizes to the **Hessian** matrix

$$\nabla^2 f(\theta) = \left[\frac{\partial^2 f}{\partial\theta_i \partial\theta_j}\right]_{i,j=1}^{p}.$$

At a local minimum of a smooth objective, the Hessian is positive definite. For MLE, the large-sample covariance matrix is approximated by the inverse curvature of the negative log-likelihood:

$$\mathrm{Var}(\hat{\theta}) \approx \left[-\nabla^2 \ell(\theta)\big|_{\theta=\hat{\theta}}\right]^{-1}.$$

This is a full covariance matrix, not just a list of variances. Off-diagonal entries matter because estimates of different parameters can be strongly correlated.

4.13 Numerical Hessian and Gamma example

A representative implementation uses `numdifftools` to approximate the Hessian numerically:

```
import numdifftools as nd
Hess = nd.Hessian(neg_loglike)
H = Hess(hat_theta, data)
cov_hat = np.linalg.inv(H)
```

In the Gamma example, the estimated covariance matrix has substantial positive covariance between $\hat{\alpha}$ and $\hat{\beta}$. Thus separate one-dimensional confidence intervals do not describe the full joint uncertainty.

The scatter/ellipse example here shows repeated MLE pairs lying along a narrow positively sloped cloud. A candidate pair can lie inside each separate marginal 95% interval but still be implausible jointly because it lies far away from this correlated direction. This also matters for functions of the parameters such as

$$\mathbb{E}(X) = \frac{\alpha}{\beta}:$$

uncertainty in the ratio depends on the *joint* behavior of $\hat{\alpha}$ and $\hat{\beta}$, including their covariance.

4.14 Fisher information, asymptotic bias, and Cramér–Rao bound

The observed curvature used above depends on the realized data. In sufficiently simple models, one can instead derive its expected value under repeated samples. This expected curvature is the **Fisher information matrix**, often denoted $I_n(\theta)$.

A key point is three large-sample properties:

1. MLEs need not be exactly unbiased in finite samples, but under regularity conditions they are **asymptotically unbiased**: their bias converges to zero as n increases.
2. For an unbiased estimator of θ, the **Cramér–Rao lower bound** gives
$$\mathrm{Var}(\hat{\theta}) \succeq I_n(\theta)^{-1}$$
in the scalar case meaning $\mathrm{Var}(\hat{\theta}) \geq I_n(\theta)^{-1}$.
3. Combined with asymptotic unbiasedness and the inverse-information variance, the MLE is asymptotically efficient: it approximately attains the smallest variance/MSE allowed by this bound among regular estimators.

4.15 Estimating functions of parameters and MLE invariance

Often the parameter vector itself is not the final quantity of interest. If the fitted model has parameter θ, we may instead want a function $g(\theta)$. For the Gamma parameterization used earlier, $\theta = (\alpha, \beta)$ but two natural targets are

$$g_1(\alpha, \beta) = \frac{\alpha}{\beta}, \qquad g_2(\alpha, \beta) = \frac{\alpha}{\beta^2},$$

the mean and variance. In risk management, the target may instead be a tail probability or a risk measure.

Maximum likelihood has the **invariance property**:

$$\hat{\theta} \text{ is the MLE of } \theta \quad \Longrightarrow \quad g(\hat{\theta}) \text{ is the MLE of } g(\theta).$$

Equivalently, if the probability model is reparameterized in terms of $g(\theta)$ and the likelihood is maximized in that parameterization, the fitted probability model is unchanged.

For example, if $X_1, \ldots, X_n \overset{iid}{\sim} \text{Exponential}(\lambda)$, then $\hat{\lambda} = 1/\bar{X}$. For $a > 0$,

$$\Pr(X_i > a) = e^{-\lambda a},$$

so invariance immediately gives

$$\widehat{\Pr(X_i > a)}_{\text{MLE}} = e^{-a\hat{\lambda}} = e^{-a/\bar{X}}.$$

Unbiasedness is not invariant under transformations. Even if $\hat{\theta}$ is unbiased for θ, $g(\hat{\theta})$ need not be unbiased for $g(\theta)$. A familiar example is that s^2 is unbiased for σ^2 under standard conditions, but $s = \sqrt{s^2}$ is not generally unbiased for σ.

4.16 Reparameterization can improve numerical optimization

The invariance property is useful computationally because we can optimize in a parameterization with easier numerical behavior and transform back afterward. For example, a variance parameter σ^2 must satisfy $\sigma^2 \geq 0$. If its true value is close to zero, an iterative optimizer can struggle near the boundary. Instead optimize over

$$\tau = \log(\sigma^2) \in \mathbb{R},$$

and recover $\sigma^2 = e^{\tau}$ after optimization.

4.17 Delta Method: one-dimensional case

The invariance property gives the point estimate $g(\hat{\theta})$. To quantify its uncertainty, use the **Delta Method**. If

$$Y_n \approx N(\mu, \sigma_n^2), \qquad \sigma_n^2 \to 0,$$

and $g'(\mu)$ exists and is nonzero, then

$$g(Y_n) \approx N\big(g(\mu),\, [g'(\mu)]^2 \sigma_n^2\big)\,.$$

The intuition is local linearization. Near μ,

$$g(x) \approx h(x) = a + g'(\mu)x,$$

the tangent line to g. As Y_n becomes concentrated near μ, $g(Y_n)$ behaves increasingly like the linear transformation $h(Y_n)$, whose variance is $[g'(\mu)]^2 \operatorname{Var}(Y_n)$.

If $X_1, \ldots, X_n$ are iid with positive support, mean μ, and variance σ^2, the CLT gives

$$\bar{X} \approx N\left(\mu, \frac{\sigma^2}{n}\right).$$

Taking $g(u) = \log u$, $g'(u) = 1/u$, so

$$\log(\bar{X}) \approx N\left(\log \mu, \frac{\sigma^2}{n\mu^2}\right).$$

4.18 Delta Method: multiple parameters

Suppose a p-parameter estimator is approximately multivariate normal,

$$\hat{\theta} \approx N_p(\theta, \Sigma),$$

and $g(\theta)$ is an m-dimensional vector with $m \leq p$. Let G be the $m \times p$ **Jacobian** matrix,

$$G_{ij} = \frac{\partial g_i(\theta)}{\partial \theta_j}.$$

Then, assuming the resulting covariance has full rank,

$$g(\hat{\theta}) \approx N_m\big(g(\theta),\ G\Sigma G^T\big)\,.$$

For a scalar target g, the approximate variance is the quadratic form $G\Sigma G^T$ and the standard error is its square root. In code, `numdifftools.Jacobian` can compute G numerically, just as `Hessian` was used for MLE covariance.

Inference logic. A test statistic compares an estimate with its null value relative to its standard error. A small p-value means the observed statistic would be unusually extreme under the null model. Statistical significance is not the same as economic importance, and non-significance can reflect either a genuinely small effect or a large standard error.

5 Part V — Evaluating the Fitted Model

Evaluation has three different targets that should never be mixed. Estimator quality concerns bias, variance, standard error, MSE, and information. In-sample fit concerns RSS, R^2, and maximized likelihood. Generalization concerns performance on new data. A procedure can score well on one target and poorly on another.

5.1 Three evaluation targets

For regression with an intercept,

$$\mathrm{TSS} = \mathrm{ESS} + \mathrm{RSS}, \qquad R^2 = 1 - \frac{\mathrm{RSS}}{\mathrm{TSS}}.$$

These summarize the observed training sample. They do not establish that the functional form is correct or that the model will generalize. Likewise, a sharply estimated coefficient does not prove that the chosen model class is adequate.

5.2 Coefficient of determination

With an intercept,

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} = 1 - \frac{\mathrm{RSS}}{\mathrm{TSS}}, \qquad 0 \le R^2 \le 1.$$

In simple linear regression,

$$R^2 = r_{xy}^2.$$

R^2 is the fraction of response sum-of-squares explained relative to an intercept-only model.

R^2 is overused. A high R^2 does not show that a straight-line model has the correct shape; a nonlinear relation can have very high R^2. A low R^2 does not automatically make a correctly specified noisy model useless. Use residual diagnostics to assess fit shape; if the model is well specified, R^2 then helps describe predictive/explanatory power.

5.3 Anscombe's quartet: numerical summaries are not enough

Anscombe's quartet consists of data sets with the same or nearly the same simple summary statistics and the same fitted regression line, but dramatically different scatter plots. Therefore coefficients, R^2, standard errors, and other numerical summaries do not tell the whole story.

Always visualize the fit. Diagnostic plots can reveal nonlinearity, heteroskedasticity, outliers, and leverage patterns hidden by a regression summary table.

5.4 Duplicating every observation: pseudo-replication

If every observed pair (x_i, y_i) is duplicated:

- the fitted line and $(\hat{\beta}_0, \hat{\beta}_1)$ do not change;
- R^2 does not change because both RSS and TSS double;
- the software treats the sample size as doubled, so the estimated standard errors shrink (roughly by $\sqrt{2}$);
- $|t|$ statistics inflate, potentially making effects look spuriously significant.

This illustrates why statistical inference requires genuinely independent information, not copied rows.

5.5 What can happen when a predictor is added?

For

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \varepsilon,$$

adding a predictor can cause previously included predictors to:

- switch from significant to non-significant;
- switch from non-significant to significant;
- change sign.

Three mechanisms emphasized here:

1. If X_1 and X_2 are strongly correlated, variation previously attributed to X_1 may be reassigned to X_2; the partial effect of X_1 can shrink or become unstable.
2. If X_1 and X_2 are nearly uncorrelated but X_2 explains substantial residual variation in Y, then $\hat{\beta}_1$ may barely move while $\hat{\sigma}^2$ and $\mathrm{SE}(\hat{\beta}_1)$ fall, increasing $|t|$.
3. **Simpson's paradox:** an overall/marginal relationship can have the opposite sign from within-group/conditional relationships.

With two predictors, OLS slopes can be written as

$$\hat{\beta}_1 = \frac{r_{y1} - r_{y2} r_{12}}{1 - r_{12}^2} \frac{s_y}{s_1}, \qquad \hat{\beta}_2 = \frac{r_{y2} - r_{y1} r_{12}}{1 - r_{12}^2} \frac{s_y}{s_2}.$$

These formulas make the role of predictor correlation explicit.

6 Part VI — Selecting Model Complexity

Additional flexibility mechanically improves or preserves training fit: adding regression predictors cannot increase training RSS, and enlarging a nested probability family cannot lower its maximized likelihood. Selection therefore asks whether the improvement in fit is worth the additional complexity.

6.1 Why predictor selection is needed

With many candidate predictors, we want a model that generalizes rather than one that merely minimizes training RSS. Training R^2 cannot be used by itself because it never decreases when predictors are added.

6.2 Akaike Information Criterion (AIC)

For linear regression, AIC can be represented (up to constants and monotone transformations that do not affect the model ordering) as

$$\mathrm{AIC} = n \log(\mathrm{RSS}) + 2k,$$

where k is the total number of fitted coefficient parameters in the candidate model. Software often reports an equivalent form such as $n \log(\mathrm{RSS}/n) + 2k$ plus constants. The important idea is unchanged:

- fit term: lower RSS is better;
- complexity penalty: more parameters are penalized;
- choose the model with the **smallest AIC**.

AIC has a theoretical motivation under a probabilistic model; if the assumed error distribution is badly wrong, that motivation can weaken.

6.3 Polynomial forward-rate example and scaling

A representative implementation constructs approximate instantaneous forward rates from zero-coupon prices. If

$$P(T) = F \exp\left(-\int_0^T r(t)\,dt\right),$$

then

$$r(T) = -\frac{\partial}{\partial T}\log P(T).$$

On a maturity grid,

$$Y_i \approx -\frac{\log P(T_i) - \log P(T_{i-1})}{T_i - T_{i-1}},$$

and one can model $Y_i = r(T_i) + \varepsilon_i$.

A polynomial mean model takes

$$Y_i = \sum_{j=0}^{q} \beta_j T_i^j + \varepsilon_i.$$

Before creating high powers, maturity is scaled to mean zero and variance one. Otherwise large raw values of T^j can create severe numerical-conditioning problems.

> Individual t-tests are not a safe variable-selection rule for correlated polynomial terms. Each t-test asks whether one coefficient matters *conditional on all other included terms*; correlated terms should often be assessed as a collection.

6.4 Exhaustive search

With p candidate predictors, an exhaustive subset search compares 2^p subsets (if the intercept is always kept). This is feasible only for relatively small p.

A representative implementation uses `itertools.combinations` to generate subsets, fit every candidate model, and choose the one minimizing AIC.

6.5 Stepwise regression

When exhaustive search is infeasible, a stepwise procedure can be used:

- **forward selection:** start small and add one predictor at a time;
- **backward selection:** start with the full model and remove one predictor at a time.

At each step, take the move that reduces AIC the most; stop when no move improves AIC.

> Stepwise procedures are greedy. They are computationally useful but do *not* guarantee the globally AIC-optimal subset.

6.6 Adjusted R^2, BIC, and Mallows' C_p

Adjusted R^2 penalizes the ordinary R^2 for the number of predictors:

$$R^2_{\text{adj}} = 1 - \frac{(1-R^2)(n-1)}{n-p-1}.$$

A useful practical point is that that this is a relatively weak complexity penalty and generally do not recommend it as the primary selection criterion.

The Bayesian Information Criterion (Schwarz criterion) is

$$\text{BIC} = n\log\left(\frac{\text{RSS}}{n}\right) + p\log n.$$

BIC penalizes complexity more strongly than AIC. Under suitable assumptions, if the true model is among the candidates, BIC selects it with probability tending to one as n grows.

Mallows' C_p is

$$C_p = \frac{\text{RSS}}{\hat{\sigma}^2} - n + 2p.$$

It estimates expected prediction error by correcting the optimistic/downward-biased training residual error. For large n, C_p and AIC are closely related.

6.7 Leave-one-out cross-validation and PRESS

In leave-one-out cross-validation (LOOCV), fit the model n times, each time leaving out one observation. Let $\hat{y}_{i(-i)}$ be the prediction for observation i from the model trained without i. Define

$$\text{PRESS} = \sum_{i=1}^{n} \left(y_i - \hat{y}_{i(-i)}\right)^2.$$

PRESS is the Prediction Error Sum of Squares.

Usually

$$\text{PRESS} \geq \text{RSS},$$

because ordinary fitted values are allowed to use observation i itself and are therefore optimistic for prediction.

For linear regression, LOOCV does not require n refits. If $e_i = y_i - \hat{y}_i$ and h_{ii} is the leverage,

$$y_i - \hat{y}_{i(-i)} = \frac{e_i}{1 - h_{ii}}, \qquad \text{PRESS} = \sum_{i=1}^{n} \left(\frac{e_i}{1 - h_{ii}}\right)^2.$$

High-leverage points receive a large correction because deleting them can change the fit substantially.

6.8 Trace of the hat matrix as model complexity

For a projection matrix H,

$$\text{tr}(H)$$

measures model complexity and is often interpreted as an equivalent/effective number of degrees of freedom.

Two useful extremes are:

- intercept-only model: H has all entries $1/n$ and $\text{tr}(H) = 1$;
- perfectly interpolating model: $H = I$, so $\hat{\mathbf{y}} = \mathbf{y}$ and $\text{tr}(H) = n$.

For standard OLS with p predictors plus an intercept, $\text{tr}(H) = p + 1$.

6.9 AIC versus PRESS

AIC and PRESS often give similar model choices, but they represent different ideas.

PRESS advantages: direct estimate of out-of-sample prediction error; weaker distributional assumptions. **PRESS disadvantages:** it can have substantial sampling variance, especially for small/moderate n; LOOCV training sets are very similar to the full sample.

AIC advantages: typically more stable and sometimes easier to compute in more general settings. **AIC disadvantages:** its theory depends on specifying a likelihood/error distribution.

6.10 LASSO

A regularized alternative is LASSO:

$$\hat{\boldsymbol{\beta}}_{\text{LASSO}} = \arg\min_{\boldsymbol{\beta}} \left\{ \sum_{i=1}^{n} (y_i - x_i^T \boldsymbol{\beta})^2 + \lambda \sum_{j=1}^{p} |\beta_j| \right\}.$$

The L_1 penalty shrinks coefficients toward zero and can set some coefficients exactly to zero, producing variable selection and shrinkage simultaneously. The penalty parameter λ is commonly chosen by cross-validation.

6.11 Why likelihood alone is not enough

Likelihood values can be compared across candidate probability models when they are fitted to **exactly the same observed data**. However, simply choosing the model with the largest maximized likelihood encourages overfitting: a more flexible model has more ways to fit sample-specific irregularities.

The objective is not to fit the observed sample perfectly, but to choose a model that represents the underlying population well. A feature visible in one finite sample may be only an **artifact** of sampling variability.

6.12 Fitting the four return models by maximum likelihood

The four candidates are Normal, GED, nonstandard t, and skewed t. the examples multiply returns by 100 before fitting because the raw daily returns are small enough to create numerical problems. The distribution `.fit(...)` methods use maximum likelihood.

For the PG data, the maximized log-likelihoods are

Model	maximized log-likelihood
Normal	−4346.11
GED	−4023.61
Nonstandard t	−3982.99
Skewed t	−3980.38

The more flexible models attain larger log-likelihoods, but these numbers should *not* be ranked naively because the models have different numbers of parameters.

6.13 Nested models and overfitting

Model 1 is **nested** in Model 2 if Model 1 is a special case of Model 2. Then the maximized likelihood under Model 2 must be at least as large as under Model 1, because Model 2 can always choose the parameter values that reproduce Model 1.

In this example:

- Normal is nested in GED ($\beta = 2$);
- nonstandard t is nested in skewed t ($a = b$).

Therefore the larger likelihood of the more flexible model is partly automatic and is not, by itself, evidence of better population fit.

A useful illustration considers this with $n = 50$ observations drawn from a true Uniform$(0, 1)$ distribution. A finite-sample histogram is not perfectly flat. Fitting a Beta(α, β) distribution can chase random bumps and dips in the histogram. Those apparent "features" are artifacts, so the fitted curve is an example of overfitting.

6.14 General AIC for probability models

AIC balances likelihood fit against model complexity:

$$\text{AIC} = -2\ell(\hat{\theta}) + 2k,$$

where k is the total number of fitted free parameters. **Smaller AIC is preferred.** The term $-2\ell(\hat{\theta})$ rewards fit and $2k$ penalizes flexibility. The regression form encountered earlier is a special case of this general likelihood criterion.

For the four PG return models, the parameter counts are 2 (Normal), 3 (GED), 3 (nonstandard t), and 4 (skewed

t). The resulting AIC values are

Model	AIC
Normal	8696.22
GED	8053.22
Nonstandard t	7971.97
Skewed t	7968.75

so AIC selects the **skewed** t. Its density can look nearly identical to the nonstandard t around the center while differing materially in the tails; the ratio of the two fitted PDFs makes that tail difference visible.

Likelihood versus AIC. Maximized likelihood never penalizes extra parameters. AIC does. When comparing models of different complexity on the same data, do not interpret the largest raw likelihood as automatically best.

6.15 Selecting a GARCH model by AIC

The earlier GARCH(1,1) idea is extended to a family indexed by orders (p, q) and by the innovation distribution:

$$X_t = \mu + \sigma_t \epsilon_t,$$

$$\sigma_t^2 = \omega + \sum_{i=1}^{p} \alpha_i X_{t-i}^2 + \sum_{j=1}^{q} \beta_j \sigma_{t-j}^2,$$

where the ϵ_t are iid with mean zero, variance one, and a chosen distribution such as Normal, GED, t, or skewed t. The candidate set therefore varies both **dynamic complexity** (p, q) and **distributional shape**.

The AIC grid search compares iid/constant-volatility models and GARCH orders up to 2 for each innovation distribution. The smallest reported AIC is

$$\boxed{\text{GARCH}(1,1) \text{ with skewed-}t \text{ innovations: AIC } = 7737.47.}$$

For the same GARCH(1,1) skewed-t specification, forcing the mean to zero gives AIC ≈ 7739.25. Because lower AIC is better, AIC slightly prefers keeping the fitted constant mean μ.

Model-selection logic. Better in-sample fit is useful only after charging for the flexibility used to obtain it. For probability models, AIC operationalizes this as

$$\text{fit cost } (-2\ell) + \text{ complexity penalty } (2k).$$

The same principle applies when choosing a return distribution, a GARCH order, or both jointly.

7 Part VII — Diagnosing Model Failure

Good numerical fit is not enough. Diagnostics test whether the fitted structure and assumptions remain credible. Residual plots target mean shape and variance structure; Q–Q plots target distributional shape; VIF targets predictor redundancy; leverage and Cook's distance target unusual or influential observations.

7.1 Residual diagnostics

The most useful basic plots are:

1. residuals versus fitted values;
2. residuals versus each predictor;
3. normal probability (Q–Q) plot of residuals.

If the mean model and error assumptions are suitable, residual plots should look like random scatter around zero, with no systematic curvature or changing spread.

An important algebraic fact is

$$\text{Corr}(\hat{\boldsymbol{\varepsilon}}, \hat{\mathbf{y}}) = 0$$

when an intercept is included (more precisely, their sample covariance is zero). This follows because residuals are orthogonal to fitted values. Therefore "zero residual–fitted correlation" does *not* prove the model is correct; the plot can still contain a strong nonlinear pattern.

7.2 Normal Q–Q plots

A Q–Q plot compares ordered residuals with expected order statistics from a standard normal distribution. If the points lie roughly on a straight line, normality is plausible.

Typical tail patterns:

- **Heavier than normal tails:** both tails bend away from the reference line in the direction of more extreme observed values;
- **Lighter than normal tails:** observed extremes are less extreme than normal-theory quantiles.

The comparison is valid even if the residuals do not have unit variance because linear rescaling of a normal variable remains normal.

Why normal residuals are useful:

- OLS becomes the MLE;
- OLS estimators are exactly normal;
- standardized estimators have exact t distributions;
- least squares is particularly appropriate when the error distribution is symmetric and lacks extreme tails.

For large n, asymptotic normality of OLS can still hold under weaker conditions, but exact small-sample inference needs stronger assumptions.

7.3 Multicollinearity

Multicollinearity occurs when one or more predictors are strongly linearly related to the others. In extreme cases $X^T X$ becomes nearly singular and OLS coefficients are numerically unstable.

Consequences:

- large coefficient variances and standard errors;
- coefficients can change strongly when correlated predictors are added or removed;
- individual predictor contributions become difficult to distinguish even when the combined predictive signal is strong.

Importantly, multicollinearity does *not* require one predictor to have high pairwise correlation with another. A predictor can be well predicted by a *linear combination* of several others.

7.3.1 Variance inflation factor (VIF)

To assess predictor X_k, regress it on all the other predictors and compute the resulting R_k^2. Then

$$\mathrm{VIF}_k = \frac{1}{1 - R_k^2}.$$

A large VIF signals multicollinearity. In OLS,

$$\mathrm{Var}(\hat{\beta}_k) = \mathrm{VIF}_k \times \left[\mathrm{Var}(\hat{\beta}_k) \text{ if } X_k \text{ were the only predictor}\right].$$

So VIF directly measures how much other predictors inflate the variance of $\hat{\beta}_k$.

the examples do not give a universal cutoff or automatic deletion rule. A pair of correlated variables may still be retained if one has a much stronger substantive/response relationship. Principal components, ridge regression, and LASSO are possible strategies for larger correlated predictor sets.

7.4 Influential observations and Cook's distance

An observation is influential if deleting it would substantially change the fitted model. Influence combines two ingredients:

- a large residual (response outlier relative to the fitted model);
- high leverage (unusual predictor location).

Cook's distance for observation j can be defined by the aggregate change in fitted values when j is removed:

$$D_j = \frac{\sum_{i=1}^{n}(\hat{y}_i - \hat{y}_{i(-j)})^2}{(p+1)\hat{\sigma}^2}.$$

For OLS, this can be calculated without refitting:

$$D_j = \frac{\hat{e}_j^2}{(p+1)\hat{\sigma}^2}\frac{h_{jj}}{(1-h_{jj})^2}.$$

Large residual magnitude and/or large leverage can therefore create large Cook's distance.

the examples explicitly avoid a single reliable hard cutoff: inspect the Cook's distances and identify observations that are extreme relative to the rest.

7.4.1 How to handle an influential point

Do *not* arbitrarily delete influential observations. Instead:

1. Check whether the point is a data error. Correct it or, if impossible and demonstrably erroneous, remove it.
2. Check whether the assumed linear relationship is inadequate; a nonlinear model may fit the responsible feature more flexibly.
3. Ask whether the point represents a genuinely different population/regime that should be modeled separately.
4. Transform predictors/responses to pull extreme values inward.
5. Consider WLS or robust regression.

8 Part VIII — Repairing the Model

A remedy should target the diagnosed failure. Changing standard errors does not repair a wrong mean function; robust fitting is not the same as heteroskedasticity-consistent inference; and transforming a response changes a different part of the model than transforming a predictor.

8.1 A. Repairing mean-structure misspecification

Systematic curvature or group-dependent effects indicate that the conditional mean is too restrictive. Available repairs include additional predictors, nonlinear or polynomial terms, indicator coding, interactions, or an alternative domain-specific mean model. These changes modify m_θ and therefore require refitting and re-diagnosis.

8.2 B. Outliers and heavy tails

Outliers can arise from data problems, a target population with genuinely extreme observations, or an error distribution with heavier tails than assumed. Robust losses and heavy-tailed probability models address different versions of this problem.

8.3 Robust regression and Huber loss

OLS minimizes squared residuals. Squaring grows rapidly, so one extreme residual can exert enormous influence. This is efficient under normal errors but not robust to heavy tails/outliers.

Huber loss is

$$\rho_{\mathrm{Hub}}(x) = \begin{cases} x^2, & |x| < k, \\ k(2|x| - k), & \text{otherwise}, \end{cases} \qquad k > 0.$$

The default value in the software example is $k = 1.345$.

Near zero, Huber behaves like squared error; in the tails, it grows approximately linearly and therefore downweights

extremes. Large k approaches least squares; very small k approaches L_1-type behavior (up to an irrelevant scaling of the loss).

Robust regression is useful when the error distribution may be non-normal/heavy-tailed and extreme observations are a concern. In the Fama–French example, robust regression reduces the impact of the most extreme residuals and gives a slightly different fit.

Python cue. `statsmodels.RLM` with a Huber norm implements robust linear modeling.

8.4 C. Heteroskedasticity: three distinct repair strategies

Heteroskedasticity concerns the scale component rather than necessarily the conditional mean. The important distinction is whether we change the response/model, change fitting weights, or keep OLS coefficients and only repair the covariance estimate.

8.5 Heteroskedasticity

Homoskedasticity assumes common error variance. **Heteroskedasticity** means the conditional error variance changes across observations. A residual-vs-fitted plot may show a fan/funnel or changing-width pattern.

Why it matters: ordinary OLS coefficient estimates can remain unbiased under suitable mean-model assumptions, but the usual standard errors, confidence intervals, and hypothesis tests are wrong because their variance formulas assume common σ^2.

Three broad remedies are available. three broad remedies.

8.5.1 1. Transform the response

If error variance increases with the response magnitude, transformations such as

$$\log(Y) \quad \text{or} \quad \sqrt{Y}$$

can "pull in" large values and stabilize variance.

8.5.2 2. Weighted least squares and feasible GLS

If $\mathrm{Var}(\varepsilon_i) = \sigma_i^2$ is known, use WLS with

$$w_i = 1/\sigma_i^2.$$

When the variances are unknown, feasible generalized least squares (FGLS) can be used:

1. fit an initial unweighted least-squares model;
2. model $\log(\hat{e}_i^2)$ using the predictors;
3. use fitted values from this variance model to estimate σ_i^2;
4. refit by WLS using weights $1/\hat{\sigma}_i^2$.

A practical caution is that FGLS tends to be more reliable with large samples and can perform poorly with smaller ones.

8.5.3 3. Heteroskedasticity-consistent (sandwich) standard errors

Keep the OLS coefficient estimates but replace the covariance formula. If

$$\Sigma = \mathrm{Var}(\boldsymbol{\varepsilon}),$$

then

$$\mathrm{Var}(\hat{\boldsymbol{\beta}}) = (X^T X)^{-1} X^T \Sigma X (X^T X)^{-1}.$$

Estimate Σ from residuals. The HC3 choice used here estimates its diagonal entries as

$$\hat{\Sigma}_{ii} = \frac{\hat{e}_i^2}{(1 - h_{ii})^2}.$$

This adjusts standard errors for heteroskedasticity and leverage while leaving the OLS point estimates unchanged.

8.6 D. Multicollinearity responses

When predictors overlap strongly, individual coefficient attribution can become unstable even when their combined predictive signal remains useful. The available responses include retaining substantively necessary variables, redesigning the predictor set, or using shrinkage/dimension-reduction approaches such as LASSO, ridge, or PCA when appropriate to the objective.

8.7 E. Power transformations and extreme scales

8.8 Motivation

Power transformations are used to reduce skewness/extreme predictor values and to make regression residuals closer to normal and/or homoskedastic. The goal is not "normality for its own sake"; normal-like predictors reduce extreme leverage, and a transformed response can make the regression error assumptions more realistic.

8.9 Box–Cox transformation

For a strictly positive response $y > 0$,

$$y^{(\lambda)} = \begin{cases} \dfrac{y^\lambda - 1}{\lambda}, & \lambda \neq 0, \\ \log y, & \lambda = 0. \end{cases}$$

This single family includes transformations corresponding approximately to y^2, $\sqrt{y}$, $\log y$, $1/\sqrt{y}$, etc. The special definition makes the transformation continuous in λ because

$$\lim_{\lambda \to 0} \frac{y^\lambda - 1}{\lambda} = \log y.$$

The optimal λ can be chosen by maximum likelihood, and software can report a confidence interval for it. If a simple interpretable value (e.g. $\lambda = 1/2$ for square root) lies comfortably inside the likelihood interval, the examples suggest that choosing the simpler transformation can be reasonable.

8.10 Box–Cox for regression is not the same as normalizing Y alone

The original Box–Cox regression problem is

$$Y^{(\lambda)} = \beta_0 + \sum_{j=1}^{p} \beta_j X_j + \varepsilon.$$

The best λ should be selected jointly with the regression model so that the *residuals* fit the Gaussian regression assumption well. Merely choosing λ to make the marginal distribution of Y normal ignores the predictors and can choose the wrong transformation for regression.

8.11 Yeo–Johnson transformation

Box–Cox requires strictly positive data. Yeo–Johnson generalizes the power family to all real responses:

$$y^{(\lambda, YJ)} = \begin{cases} \dfrac{(y+1)^\lambda - 1}{\lambda}, & \lambda \neq 0,\ y \geq 0, \\ \log(y+1), & \lambda = 0,\ y \geq 0, \\ -\dfrac{(-y+1)^{2-\lambda} - 1}{2-\lambda}, & \lambda \neq 2,\ y < 0, \\ -\log(-y+1), & \lambda = 2,\ y < 0. \end{cases}$$

As with Box–Cox, λ can be chosen by likelihood.

Python cues. `scipy.stats.boxcox` finds/transforms Box–Cox data; `scipy.stats.yeojohnson` is the real-valued generalization.

8.12 MSFT spread–volume example: full diagnostic workflow

A representative financial example studies intraday price spread versus trading volume and demonstrates how multiple diagnostics connect:

1. Raw spread versus volume shows strong heteroskedasticity in residuals.
2. Box–Cox regression for spread favors a λ near zero, motivating **log spread**.
3. Residual spread improves after transforming the response, though influential points can remain.
4. Raw volume is strongly right-skewed; a log transform makes its distribution much more symmetric.
5. A model using both **log spread** and **log volume** gives a substantially cleaner residual cloud.
6. The Q–Q plot becomes close to linear, supporting the normal-error approximation.
7. Cook's distances become less dominated by one or two observations; transformation can reduce leverage/influence caused by extreme raw scales.

This is the intended modeling sequence: diagnose → transform/remedy → refit → re-diagnose.

9 Part IX — Prediction

Once a model has been fitted, assessed, selected, and repaired, prediction separates uncertainty about the conditional mean from uncertainty about a new individual outcome. The latter contains irreducible future noise and therefore cannot collapse even when parameter estimation becomes extremely precise.

9.1 Point prediction

For a new simple-regression predictor value x^*,

$$\tilde{y}^* = \hat{\beta}_0 + \hat{\beta}_1 x^*$$

is the point prediction.

9.2 Prediction error variance

For a new response y^* generated under the model,

$$\mathrm{Var}(\tilde{y}^* - y^*) = \sigma^2 + \sigma^2 \left[\frac{1}{n} + \frac{(x^* - \bar{x})^2}{(n-1)s_x^2} \right].$$

The two pieces are:

1. σ^2: **irreducible error** / random scatter of a new observation around the true regression line;
2. the bracketed term: uncertainty in the fitted coefficients $\hat{\beta}_0, \hat{\beta}_1$.

Even for a large sample, the irreducible term usually dominates. This formula also assumes the linear model itself is correctly specified; it does not include model misspecification error.

The estimated prediction standard error replaces σ^2 by $\hat{\sigma}^2$.

9.3 Prediction interval

If errors are normal, a $100(1-\alpha)\%$ prediction interval is

$$\tilde{y}^* \pm t_{\alpha/2,n-2}\,\mathrm{SE}(\tilde{y}^*),$$

with a normal critical value as a large-sample approximation.

Prediction intervals widen as x^* moves away from $\bar{x}$ because coefficient-estimation uncertainty grows in extrapolative regions.

Mean confidence interval versus prediction interval. A confidence interval for the mean response $\mathbb{E}(Y \mid x^*)$ excludes the new observation's irreducible σ^2 term and is therefore narrower. A prediction interval for a new Y^* includes both parameter uncertainty and irreducible error.

9.4 Why normality can still matter for a large prediction sample

For coefficient estimates, large-sample central-limit arguments can reduce dependence on exact normal errors. For an individual future response, however, the new irreducible error ε^* does not get averaged away. If its distribution is strongly non-normal (especially heavy-tailed), a normal-theory prediction interval can remain inaccurate even when the training sample is large. Residual Q–Q plots are therefore still important.

Statsmodels cue. `get_prediction` can return both a confidence interval for the mean response and a prediction interval for a new observation. Output columns often distinguish `mean_ci_*` from `obs_ci_*`.

10 Part X — Financial Quantities Derived from the Fitted Model

Financial analysis often ends with a nonlinear quantity derived from fitted distribution parameters. This creates a final chain: probability model → parameter estimates → parameter covariance → risk measure → uncertainty of that risk measure. MLE invariance gives the plug-in point estimate; the Delta Method supplies its approximate standard error.

10.1 Risk-management application: VaR and CVaR

Let L denote loss. Under the convention used here, **VaR at tail level** α is the smallest c such that

$$\Pr(L > c) \le \alpha.$$

Conditional Value at Risk (CVaR), also called **Expected Shortfall (ES)**, is

$$\mathrm{CVaR}_\alpha = \mathbb{E}[L \mid L > \mathrm{VaR}_\alpha].$$

If $L \sim N(\mu, \sigma^2)$ and z_α is defined by

$$\Pr(Z > z_\alpha) = \alpha, \qquad Z \sim N(0,1),$$

then

$$\mathrm{VaR}_\alpha = \mu + \sigma z_\alpha, \qquad \mathrm{CVaR}_\alpha = \mu + \sigma \frac{\phi(z_\alpha)}{\alpha},$$

where ϕ is the standard-normal density. The CVaR derivation uses

$$\frac{d}{dv}\phi(v) = -v\phi(v) \quad \Longrightarrow \quad \int_{z_\alpha}^{\infty} v\phi(v)\,dv = \phi(z_\alpha).$$

To estimate CVaR from normal data with uncertainty, the example's workflow is:

1. obtain the MLE $(\hat\mu, \hat\sigma^2)$;
2. estimate $\mathrm{Var}(\hat\mu, \hat\sigma^2)$ from the inverse Hessian;
3. compute the Jacobian of
$$g(\mu, \sigma^2) = \mu + \sqrt{\sigma^2}\,\frac{\phi(z_\alpha)}{\alpha};$$
4. use $G\hat\Sigma G^T$ for the Delta-Method variance of $g(\hat\mu, \hat\sigma^2)$, then form an approximate 95% interval as estimate ± 1.96 SE.

Core intuition. MLE chooses the model that best explains the observed data. The curvature of the log-likelihood tells us how precisely the parameter is identified. More data usually make the likelihood peak sharper; the inverse curvature then shrinks, giving smaller estimator variance.

11 Implementation and API Cues

The computational functions below are useful because each corresponds to a statistical operation in the modeling workflow; the important point is what each tool computes, not memorizing syntax in isolation.

These are the functions/packages that appear repeatedly here; the key point is what they are doing statistically and how each tool maps to the modeling workflow.

Tool	Meaning here
`yfinance.Ticker(...) .history(...)`	Download historical market data; adjusted prices account for corporate actions by default.
`scipy.stats.norm.fit`	Normal MLE fit of location and scale.
`scipy.optimize.minimize`	Numerical minimization of a user-defined RSS/objective for nonlinear mean models.
Nelson–Siegel calibration package	Generate/calibrate Nelson–Siegel or Nelson–Siegel–Svensson yield curves.
implied-volatility function	Invert an option pricing model to recover volatility from a market option price.
`statsmodels.OLS / ols.from_formula`	OLS regression; exposes coefficients, SEs, t tests, residuals, fitted values, and model summaries.
`qqplot`	Residual normal probability plot.
`PolynomialFeatures`	Construct polynomial predictor powers (after scaling for numerical stability).
`itertools.combinations`	Enumerate model subsets for exhaustive selection.
`variance_inflation_factor`	Compute predictor VIFs.
`RLM(..., HuberT())`	Robust regression with Huber loss.
`get_influence().cooks_distance`	Influence ranking via Cook's distance.
`boxcox / yeojohnson`	Power transformations; Box–Cox requires positive input, Yeo–Johnson does not.
`get_prediction`	Mean-response CI and new-observation prediction interval.
`scipy.optimize.minimize`	Minimize a user-defined negative log-likelihood; parameters are the first objective-function argument; data are passed through `args`.
`scipy.stats.gamma.logpdf / t.logpdf / norm.logpdf`	Stable building blocks for likelihoods; remember Gamma rate β corresponds to SciPy scale $1/\beta$.
`numdifftools.Hessian`	Numerically approximate the Hessian; invert the negative-log-likelihood Hessian at the MLE for an approximate covariance matrix.
`numdifftools.Jacobian`	Numerically compute the Jacobian G for multivariate Delta-Method variances $G\Sigma G^T$.
`scipy.stats.gennorm.fit / logpdf`	MLE fitting and log-likelihood evaluation for the GED/generalized-normal family.
`scipy.stats.t.fit / logpdf`	Fit/evaluate a location-scale nonstandard t model.
`scipy.stats.jf_skew_t.fit / logpdf`	Fit/evaluate the Jones–Faddy skewed-t distribution.
`arch.arch_model`	Fit GARCH families and compare p, q, innovation `dist`, mean specification, and the fitted model's `.aic`.

12 Practical Q&A and Problem-Solving Guide

The same modeling logic provides a reliable way to approach practical questions, interpretation tasks, and short analytical problems: identify the model and assumptions, perform the relevant calculation or diagnostic, state its statistical meaning, and then give the appropriate action or conclusion. The following compact guide collects reusable question-and-answer patterns for applying the framework in practice.

12.1 Question-type map

Question type	What a complete answer should contain	Key related topics
Interpret Python output / figures	State what is being estimated or plotted, read the key numerical/visual evidence, then state the statistical implication.	OLS output; residual plots; Q–Q plots; VIF; Cook's distance; AIC/BIC/PRESS; transformation diagnostics; MLE/Hessian output; GARCH AIC.
Derive a likelihood	Write the density/conditional density, form the product over observations, take logs, then maximize or explain numerical maximization.	Exponential, Normal, Gamma, heavy-tailed t, GARCH conditional likelihood, MLE invariance.
Derive an MSE	Start from $\mathbb{E}[(\hat{\theta}-\theta)^2]$, decompose into variance plus squared bias, then substitute the estimator's moments.	Sample mean, variance estimators, shrinkage, bias–variance tradeoff.
Explain steps in a situation	Diagnose the assumption/problem first, then choose a remedy that addresses that problem and explain what changes.	Heteroskedasticity, multicollinearity, nonlinearity, outliers/influence, heavy tails, overfitting, transformations, regularization.
Explain a concept	Give the definition, the mechanism/intuition, and one consequence or example from the course.	Bias/variance, overfitting, leverage/influence, nested models, AIC penalty, MLE curvature, Delta Method.
Construct / interpret a CI	Identify the estimator, its SE and reference distribution/approximation, compute the interval, and interpret repeated-sampling coverage correctly.	OLS t intervals, large-sample normal intervals, MLE Hessian SE, Delta-Method CI, prediction interval versus mean-response CI.

12.2 Interpreting Python output and figures

Use the same four-step structure almost every time:

1. **Identify the object:** coefficient table, residual plot, Q–Q plot, density fit, model-selection table, influence statistic, transformation profile, etc.
2. **Read the evidence:** coefficient/SE/t/CI, R^2, AIC, VIF, Cook's distance, shape of residuals, tail deviations, or fitted-vs-empirical discrepancy.
3. **Translate into a statistical statement:** significance/uncertainty, assumption violation, model misspecification, overfitting risk, or better/worse penalized fit.
4. **State the consequence:** keep/reject a term, change the model, use a robust SE, transform, regularize, or choose the smaller AIC/BIC/PRESS value.

Never stop at "the plot looks bad." Name the pattern, connect it to the corresponding assumption/model feature, and state what you would do next.

Common outputs.

- **OLS coefficient table:** $\hat{\beta}$ is the estimated effect conditional on the other predictors; its SE measures uncertainty; a t statistic compares the estimate with the null value; the confidence interval gives a range of parameter values compatible with the model and data.
- **R^2 versus adjusted R^2:** ordinary R^2 cannot decrease after adding predictors; adjusted R^2 can decrease because it charges for extra predictors.
- **AIC/BIC/PRESS:** smaller is preferred. Raw likelihood or training RSS alone favors more flexible models and therefore is not a fair complexity-adjusted comparison.
- **VIF:** large VIF means a predictor is well explained by the other predictors, inflating coefficient variance and making signs/SEs unstable.
- **Cook's distance / leverage:** leverage measures unusual predictor position; Cook's distance measures how much the fitted model is affected by an observation.

Common figures.

- **Residual vs fitted:** random cloud with roughly constant spread is desired. Curvature suggests mean-model misspecification; a funnel suggests heteroskedasticity; isolated extreme points suggest outliers/influence.
- **Normal Q–Q:** approximate straight line supports normality. Systematic tail bending means the tails differ from Normal; strong two-sided deviations in returns indicate heavier tails.
- **Density on a log scale:** emphasizes relative differences in very small tail probabilities. This is why the PG return example reveals Normal-tail underfitting clearly.
- **Model-fit comparison:** ask whether the improvement is in the center or tails and whether the extra flexibility is being penalized. For nested models, higher maximized likelihood is partly automatic.

12.3 Deriving a likelihood function

For independent observations $X_1, \ldots, X_n$ from density/pmf $f(x \mid \theta)$, the standard derivation is

$$L(\theta) = \prod_{i=1}^{n} f(x_i \mid \theta), \qquad \ell(\theta) = \log L(\theta) = \sum_{i=1}^{n} \log f(x_i \mid \theta).$$

Then:

1. simplify $\ell(\theta)$;
2. differentiate and solve the score equation when a closed form exists;
3. check parameter constraints and that the solution is a maximum;
4. if no closed form exists, numerically minimize $-\ell(\theta)$;
5. for uncertainty, use the inverse negative Hessian at the MLE.

Representative examples.

- **Exponential:** the product likelihood becomes a power of the rate times an exponential in $\sum X_i$, and the MLE is obtained from the score equation.
- **Normal errors:** maximizing likelihood in a regression with iid Normal errors is equivalent to minimizing RSS for the mean parameters.
- **Gamma / t / GED / skewed t:** the same likelihood principle applies even when optimization is numerical.
- **GARCH:** observations are not iid unconditionally. Use the product of *conditional* densities, where each σ_t^2 is generated recursively from the past.

Likelihood is a function of the parameter after the data are observed. The data values are fixed in $L(\theta)$; θ varies. Also distinguish the likelihood value used for estimation from a probability statement about the parameter.

If the desired parameter is $g(\theta)$ rather than θ itself, MLE invariance gives the point estimate $g(\hat{\theta})$. It does *not* automatically give its SE; use the Delta Method/Jacobian for uncertainty.

12.4 Deriving MSE and explaining bias–variance tradeoff

Start from the definition and add/subtract $\mathbb{E}(\hat{\theta})$:

$$\hat{\theta} - \theta = \big(\hat{\theta} - \mathbb{E}\hat{\theta}\big) + \big(\mathbb{E}\hat{\theta} - \theta\big).$$

Squaring and taking expectations gives

$$\mathrm{MSE}(\hat{\theta}) = \mathbb{E}[(\hat{\theta} - \theta)^2] = \mathrm{Var}(\hat{\theta}) + \mathrm{Bias}(\hat{\theta})^2.$$

The cross term is zero because $\mathbb{E}[\hat{\theta} - \mathbb{E}(\hat{\theta})] = 0$.

How to use this framework:

1. compute $\mathbb{E}(\hat{\theta})$ and hence the bias;
2. compute $\mathrm{Var}(\hat{\theta})$;
3. substitute both into variance + squared bias;
4. compare estimators using MSE, not unbiasedness alone.

The worked examples show why this matters. $\bar{X}$ is unbiased with variance σ^2/n. A constant estimator can have zero variance but nonzero bias. For Normal variance estimation, the unbiased estimator, the MLE, and the MSE-optimal shrinkage estimator are different. Thus **unbiased does not mean best under MSE**.

Bias–variance concept answer: increasing model flexibility usually lowers approximation bias but increases estimation variance because more parameters/features are learned from the same sample. More data mainly reduce variance; they do not automatically remove structural bias from a misspecified model. Overfitting is the practical result when variance/flexibility is used to fit sample artifacts rather than population structure.

12.5 Situation questions: diagnose first, then choose the remedy

Situation	What it means	Recommended response
Heteroskedastic residuals	Error variance changes with the fitted value/predictors. OLS coefficients can still target the mean model, but usual homoskedastic SEs are unreliable.	Diagnose with residual spread. If variance structure is modeled, use WLS/FGLS; if inference is the main concern, use HC3 robust SE; a response transformation may stabilize variance.
Multicollinearity	Predictors contain overlapping information; coefficient variance is inflated.	Inspect correlations/VIF, avoid redundant variables when substantively justified, and consider regularization/dimension reduction when prediction is the goal.
Nonlinear residual pattern	Conditional mean is misspecified.	Add an appropriate nonlinear term/interaction or transform/redefine the mean model; robust SE alone does not repair a wrong mean function.
Heavy-tailed / non-normal residuals	Normal-tail approximation is poor; extreme observations are more common than expected.	Check Q–Q/tails; use robust regression or a more appropriate probability model; be cautious with normal-tail intervals/risk measures.
Influential observation	One point materially changes the fitted model.	Check leverage and Cook's distance, verify the data, determine whether the point belongs to the target population, then consider transformation/robust methods/model revision rather than deleting automatically.
Too many predictors / training fit keeps improving	Risk of overfitting.	Use adjusted R^2, AIC/BIC, PRESS/CV, LASSO, or a principled search procedure; do not select by training RSS or raw likelihood alone.
Categorical predictor coded as numbers	Numeric coding may impose a false equal-spacing/linear effect.	Use indicator variables with a baseline category; include interactions when category effects depend on another predictor.
Need prediction for a new observation	New outcome contains both parameter uncertainty and new random noise.	Use a prediction interval, which is wider than the confidence interval for the conditional mean.

Situation-question template. (1) Name the violated assumption/problem. (2) Explain how you diagnosed it. (3) State what quantity is affected—coefficients, SEs, prediction, or tail probability. (4) Choose a remedy that targets that failure. (5) State what you would check after refitting.

12.6 Explaining concepts clearly

For a concept question, a strong short answer has three layers: **definition → mechanism → consequence**.

- **Bias:** $\mathbb{E}(\hat{\theta}) - \theta$; systematic error of an estimation procedure; high bias means repeated fits are centered away from the truth.
- **Variance:** $\mathrm{Var}(\hat{\theta})$; sampling instability; high variance means repeated samples give widely different estimates.
- **Overfitting:** fitting sample-specific noise/artifacts; training fit improves while population/out-of-sample perfor-

mance may worsen.

- **Leverage vs influence:** leverage is unusual X position; influence is actual impact on the fitted model. High leverage does not guarantee high influence unless the response is also discordant.
- **Nested models:** the smaller model is obtained by restricting parameters of the larger model; therefore the larger model's maximized likelihood cannot be lower.
- **AIC:** $-2\ell(\hat{\theta}) + 2k$ trades likelihood fit against the total number of fitted free parameters; choose the smaller value among models fit to the same data.
- **MLE curvature:** a sharply peaked log-likelihood means the parameter is strongly identified; inverse curvature/Hessian gives smaller approximate variance.
- **Delta Method:** locally linearizes a nonlinear function g around the parameter estimate so uncertainty in $\hat{\theta}$ can be propagated to $g(\hat{\theta})$.

12.7 Constructing and interpreting confidence intervals

The generic structure is

$$\text{estimate} \ \pm \ \text{critical value} \times \text{SE(estimate)}.$$

Before writing numbers, identify which sampling distribution/approximation is justified.

- **Simple-regression slope under Normal errors:**

$$\hat{\beta}_1 \pm t_{n-2,\,1-\alpha/2}\ \mathrm{SE}(\hat{\beta}_1).$$

- **Large-sample approximately Normal estimator/MLE:**

$$\hat{\theta} \pm z_{1-\alpha/2}\ \mathrm{SE}(\hat{\theta}), \qquad z_{0.975} \approx 1.96.$$

- **MLE SE:** obtain the covariance matrix from the inverse negative Hessian; use the square root of the relevant diagonal element.
- **Transformed MLE target $g(\theta)$:** point estimate $g(\hat{\theta})$ by invariance; SE from the Delta Method, $G\hat{\Sigma}G^T$.
- **Mean-response CI versus prediction interval:** both center on the fitted value, but the prediction interval contains an extra new-observation variance term and is therefore wider.

Frequentist interpretation. After the interval is computed, do not say there is a 95% probability that the fixed parameter lies inside it. The correct statement is that the interval-producing procedure covers the true parameter in 95% of repeated samples under the model.

12.8 Core derivations and distinctions to know fluently

- $\mathrm{MSE} = \mathrm{Var} + \mathrm{Bias}^2$ and why the cross term vanishes.
- $\hat{\beta}_1 = s_{xy}/s_x^2$ and $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{x}$; know what changes in multiple regression.
- $\mathrm{Var}(\hat{\beta}) = \sigma^2(X^TX)^{-1}$ and the role of X^TX in multicollinearity/precision.
- $H = X(X^TX)^{-1}X^T$, leverage h_{ii}, and PRESS residual $e_i/(1-h_{ii})$.
- Likelihood product $\rightarrow$ log-likelihood sum $\rightarrow$ score/Hessian or numerical optimization.
- MLE invariance for the point estimate versus Delta Method for the transformed SE.
- AIC/BIC/PRESS/LASSO all address overfitting, but by different mechanisms: penalized fit, estimated out-of-sample error, or coefficient shrinkage/selection.
- Robust regression changes the fitting loss; HC3 changes the estimated covariance/SE. They solve different problems.
- WLS changes the fitting weights when error variances differ; ordinary OLS with robust SE keeps the OLS coefficients but changes inference.
- Confidence interval for a mean response is not a prediction interval for a future observation.

12.9 Practical mastery checklist

You should be able to do each of the following without external reference:

1. write a model clearly, identify its parameters, and distinguish a mean model from a full probability model;
2. derive or recognize OLS estimates, SEs, t statistics, R^2, leverage, and prediction uncertainty;
3. read residual, Q–Q, density, and influence plots and connect each pattern to a specific modeling problem;
4. explain bias, variance, MSE, overfitting, shrinkage, and why more flexibility is not automatically better;
5. compare models using adjusted R^2, AIC, BIC, C_p, PRESS/CV, and LASSO with the correct direction of preference;
6. diagnose heteroskedasticity, multicollinearity, nonlinear mean structure, outliers/influence, and heavy tails and select an appropriate remedy;
7. derive a likelihood/log-likelihood and explain when numerical optimization is required;
8. obtain MLE uncertainty from likelihood curvature/Hessian and propagate it through a nonlinear function using the Delta Method;
9. distinguish confidence intervals from prediction intervals and interpret frequentist coverage correctly;
10. explain why Normal return models can fail in the tails and why GED/t/skewed-t/GARCH models are compared using penalized likelihood rather than raw fit alone.

Practical reasoning rule. For almost every problem, organize the answer as *model/assumption → calculation or diagnostic → statistical meaning → action or conclusion.* Showing that chain is more valuable than reporting an isolated formula or Python number.

13 Financial Data Science: Complete Modeling Map

This final section compresses the entire lecture into one decision system. Use it as a last-pass map after the detailed review.

13.1 Full logical workflow

Stage	Main question	Main tools	Output / next check
Model	What structure can represent the data-generating process?	Mean model, scale model, innovation distribution; linear/factor/polynomial/categori Nelson–Siegel/NSS; GARCH	A parameterized model class and assumptions.
Fit	Which parameter values best match the sample?	LS/OLS, WLS, Huber, MLE, numerical optimization, conditional GARCH likelihood	Point estimates, fitted values, residuals, maximized likelihood.
Uncertainty	How variable are estimates across samples?	SE, t inference, OLS covariance, Hessian/Fisher information, Delta Method	Covariance matrix, tests, CIs, uncertainty for derived quantities.
Diagnose	Are mean, variance, distribution, and influence assumptions credible?	Residual plots, Q–Q plots, VIF, leverage, Cook's distance, likelihood curvature	A specific identified failure or evidence that assumptions are adequate.
Select	Is better fit worth extra flexibility?	adjusted R^2, AIC, BIC, C_p, PRESS/LOOCV, exhaustive/stepwise search, LASSO	A complexity-controlled candidate model.
Repair	What change targets the diagnosed problem?	nonlinear terms/interactions, transformation, robust loss, WLS/FGLS, HC3, heavy-tailed/skewed distributions, redesign/regularization	A revised model to refit and re-diagnose.

Stage	Main question	Main tools	Output / next check
Predict	What is the future mean or a new outcome?	point prediction, mean-response CI, prediction interval; tail/risk calculations	Forecast or risk quantity with uncertainty.

13.2 Model architecture menu

Component	Question	Available structures
Mean m_θ	What is $\mathbb{E}(Y \mid \text{information})$?	Constant; simple/multiple linear; Fama–French; polynomial; categorical indicators; interactions; Nelson–Siegel/NSS.
Scale s_η	How large is conditional uncertainty?	Constant σ; observation-specific σ_i; WLS/FGLS variance model; GARCH conditional volatility.
Innovation G_ϕ	What shape remains after standardizing by the scale?	Normal; GED/generalized normal; t; skewed-t.

13.3 Fitting-method menu

Method	Fits / assumes	Core idea
Least squares / OLS	Mean model	Minimize squared residuals.
WLS	Mean model with unequal known/modeled variances	Minimize weighted squared residuals, ideally $w_i = 1/\sigma_i^2$.
Huber robust fitting	Mean model with outlier resistance	Quadratic near zero, approximately linear in tails.
MLE	Full probability model	Maximize observed-data likelihood or log-likelihood.
Numerical optimization	Nonlinear LS or likelihood without closed form	Minimize RSS or negative log-likelihood subject to constraints/parameterization.
Conditional GARCH likelihood	Dynamic scale plus innovation distribution	Recursively compute σ_t and sum conditional log densities.

Coincidence. OLS coefficient estimates equal MLE coefficient estimates when regression errors are iid, homoskedastic, and Normal.

13.4 Evaluation menu

Target	Main quantities
Estimator quality	Bias, variance, SE, MSE, Fisher information.
In-sample fit	RSS/TSS/ESS, R^2, maximized likelihood/log-likelihood.
Generalization / selection	adjusted R^2, AIC, BIC, C_p, LOOCV/PRESS, LASSO/CV.
Diagnostics	Residual plots, Q–Q plot, VIF, leverage, Cook's distance, likelihood curvature.

13.5 Problem → symptom → consequence → remedy

Problem	Typical symptom	What it threatens	Course remedy
Underfitting / high structural bias	Systematic residual pattern; overly rigid model	Mean-model accuracy and prediction	Enlarge/rethink mean class; add justified nonlinear, factor, categorical, or interaction structure.
Overfitting	Training RSS keeps falling / likelihood rises but validation worsens	Generalization	AIC/BIC/C_p/PRESS/CV; LASSO; simpler model.
Nonlinear mean misspecification	Curvature in residual-vs-fitted/predictor plot	Conditional mean, coefficients, prediction	Polynomial/nonlinear terms, interactions, transformation, alternative mean model.
Heteroskedasticity	Funnel/changing residual spread	Usual SE/tests/CIs; OLS efficiency	Transform response; WLS/FGLS; or keep OLS coefficients and use HC3 robust covariance.
Heavy tails	Q–Q tail departures; extreme residuals	Normal-tail inference/prediction; OLS sensitivity	Huber/L_1-type robust fitting; t/GED/skewed-t model.
Outliers	Large residuals	Fit stability and tail assumptions	Verify data/population; robust fitting; transformation/model revision; do not delete automatically.
Multicollinearity	High VIF, large SEs, unstable signs	Individual coefficient attribution/precision	Retain if justified; redesign; LASSO/ridge/PCA where appropriate.
High leverage	Large h_{ii} / unusual predictor location	Sensitivity of fitted value; PRESS correction	Investigate scale/design; transform extreme predictors; assess influence.
Influential observation	Large Cook's distance	Entire fitted model	Verify, assess population relevance, transform/robustify/revise rather than auto-delete.
Extreme predictor skewness	A few huge predictor values	Leverage, numerical conditioning, linearity	Log/power transform; standardize before high polynomial powers.
Poor Normal tail fit	Log-density/Q–Q tail mismatch	Tail probabilities, PI, VaR/CVaR	Compare GED/t/skewed-t using likelihood plus AIC.
Time-varying volatility	Volatility clustering	Constant-variance probability model	GARCH(p, q) scale dynamics with conditional likelihood.
Asymmetric innovations	Left/right tail behavior differs	Symmetric innovation fit and tail risk	Skewed-t rather than symmetric alternatives; compare with AIC.
Flat likelihood / weak identification	Broad likelihood peak, small Hessian curvature	Large MLE variance	Report uncertainty; inspect inverse Hessian/information; collect more information when possible.
Positive-parameter boundary issues	Optimizer approaches invalid/negative values	Numerical stability	Bounds or log reparameterization, then transform back by invariance.
Pseudo-replication	Duplicated rows shrink SEs without new information	False precision and inflated $\|t\|$	Require genuinely independent information; do not treat copies as new observations.
Selecting by raw likelihood	Larger nested model always looks at least as good	Complexity-driven overfit	Use AIC (or other generalization criteria), counting all fitted free parameters.

13.6 Prediction and inference map

Object	What uncertainty/question it answers
Parameter confidence interval	Repeated-sample uncertainty for a fixed unknown parameter.
Mean-response confidence interval	Uncertainty in $\mathbb{E}(Y \mid x^*)$ from estimated model parameters.
New-observation prediction interval	Mean-estimation uncertainty plus irreducible future innovation; therefore wider than the mean CI.
Hypothesis test	Whether an estimate is large relative to its SE under a specified null.
p-value	Tail probability of a statistic at least as extreme under the null model; not the probability that the null is true.
Delta-Method interval	Propagates a parameter covariance matrix through a nonlinear derived quantity $g(\theta)$.

13.7 Complete concept checklist

- **Modeling foundations:** model class vs fitting rule; training vs population; bias, variance, overfitting, double descent; parameter/statistic/estimator/estimate; SE, MSE, shrinkage; frequentist CI interpretation.
- **Mean architecture:** constant/simple/multiple linear models; Fama–French; polynomial terms; discrete vs categorical vs ordinal; dummy coding/reference levels; all three interaction types, conditional main effects, centering; Nelson–Siegel/NSS; option calibration and implied volatility.
- **Scale and innovation:** homoskedastic vs heteroskedastic variance; time-varying σ_t; Normal, GED/generalized normal, nonstandard t, skewed-t; scale vs SD; log returns/lognormal link; log-density tail diagnostics.
- **Fitting:** LS/OLS, L_2 vs L_1, WLS/inverse variance, nonlinear calibration, design matrix, projection geometry, hat/residual-maker matrices, leverage, orthogonality, simple-regression formulas, MLE/log-likelihood/NLL, Exponential and Gamma examples, Normal-MLE/OLS equivalence, t regression, conditional GARCH likelihood, numerical constraints/initialization.
- **Uncertainty and inference:** χ^2, Student t, Normal mean/variance results, OLS covariance, residual df, exact t inference, p-values, significance vs economic size, likelihood curvature, observed/Fisher information, Hessian, $1/n$ variance scaling, regularity, Uniform counterexample/German-tank correction, invariance, reparameterization, Delta Method/Jacobian.
- **Evaluation:** RSS/TSS/ESS, R^2 and $R^2 = r^2$ special case, Anscombe, pseudo-replication, predictor-addition effects, Simpson's paradox, model quality vs estimator quality.
- **Selection:** regression AIC, general likelihood AIC with total free-parameter count k, BIC, adjusted R^2, C_p, exhaustive 2^p search, stepwise greedy search, PRESS/LOOCV and leverage identity, $\mathrm{tr}(H)$ effective complexity, LASSO/CV, nested models, Normal/GED/t/skewed-t comparison, GARCH order and innovation-distribution selection, fitted vs zero mean.
- **Diagnostics:** residual-vs-fitted/predictor plots, Q–Q tails, VIF, leverage, outlier vs leverage vs influence, Cook's distance.
- **Repairs:** nonlinear terms/interactions/categorical coding; Huber; heteroskedastic response transform/WLS/FGLS/HC3; multicollinearity responses; Box–Cox and Yeo–Johnson; likelihood choice of λ; predictor vs response transformation; MSFT spread–volume diagnose-transform-refit-re-diagnose workflow.
- **Prediction and risk:** point prediction; mean-response CI vs PI; irreducible future error; heavy tails for individual future observations; VaR/CVaR; MLE covariance → invariance → Delta-Method risk uncertainty.

13.8 Complete formula checklist

Function	Formula	Use
Estimator quality	$\mathrm{Bias}(\hat{\theta}) = \mathbb{E}\hat{\theta} - \theta$, $\mathrm{MSE} = \mathrm{Var}(\hat{\theta}) + \mathrm{Bias}^2$	Compare estimator accuracy/precision.

Function	Formula	Use
Estimator quality	$s^2 = (n-1)^{-1}\sum(X_i - \bar{X})^2$; Normal MLE uses n; MSE-optimal shrinkage uses $n+1$ in the stated family	Variance estimation and shrinkage.
Least squares	$\mathrm{RSS}(\theta) = \sum_i [y_i - m(\theta, x_i)]^2$	Fit a mean model.
Least squares	$\mathrm{WRSS} = \sum_i w_i(y_i - m_i)^2$, $w_i = 1/\sigma_i^2$	Efficient weighting under unequal known variances.
Simple regression	$\hat{\beta}_1 = s_{xy}/s_x^2 = r_{xy}s_y/s_x$, $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{x}$	Closed-form one-predictor OLS.
Multiple regression	$\hat{\beta} = (X^TX)^{-1}X^TY$	General OLS estimate.
Projection	$H = X(X^TX)^{-1}X^T$, $\hat{Y} = HY$, $e = (I-H)Y$	Fitted values, residuals, leverage.
Regression uncertainty	$\mathrm{Var}(\hat{\beta} \mid X) = \sigma^2(X^TX)^{-1}$; $\hat{\sigma}^2 = \mathrm{RSS}/(n-p-1)$	Coefficient covariance and residual variance.
Inference	$(\hat{\beta}_j - \beta_{j,0})/\,\mathrm{SE}(\hat{\beta}_j) \sim t_{n-p-1}$ under Normal errors	Tests and coefficient CIs.
Fit statistics	$\mathrm{TSS} = \mathrm{ESS} + \mathrm{RSS}$, $R^2 = 1 - \mathrm{RSS}/\mathrm{TSS}$	In-sample explained variation.
Model selection	$\mathrm{AIC} = -2\ell(\hat{\theta}) + 2k$	Penalized likelihood; k is all free fitted parameters.
Model selection	$R^2_{adj} = 1 - (1-R^2)(n-1)/(n-p-1)$	Penalized regression fit.
Model selection	$\mathrm{BIC} = n\log(\mathrm{RSS}/n) + p\log n$, $C_p = \mathrm{RSS}/\hat{\sigma}^2 - n + 2p$	Stronger parsimony / prediction correction.
Model selection	$\mathrm{PRESS} = \sum_i [e_i/(1-h_{ii})]^2$	LOOCV prediction error without n refits.
Penalized estimation	$\arg\min_\beta\{\mathrm{RSS} + \lambda\sum_j \lvert\beta_j\rvert\}$	LASSO shrinkage and exact-zero selection.
Categorical/interaction	$\mathbb{E}(Y) = \beta_0 + \beta_1X_1 + \beta_2X_2 + \beta_3X_1X_2$; $\partial\mathbb{E}(Y)/\partial X_2 = \beta_2 + \beta_3X_1$	Conditional effect under interaction.
Prediction	$\mathrm{Var}(\tilde{y}^* - y^*) = \sigma^2 + \sigma^2[1/n + (x^* - \bar{x})^2/((n-1)s_x^2)]$	New-observation prediction uncertainty.
Diagnostics	$\mathrm{VIF}_j = 1/(1-R_j^2)$	Multicollinearity of predictor j with all others.
Diagnostics	$D_j = \frac{e_j^2}{(p+1)\hat{\sigma}^2}\frac{h_{jj}}{(1-h_{jj})^2}$	Cook influence measure.
Robust fit	$\rho_H(r) = r^2$ for $\lvert r\rvert < c$; $c(2\lvert r\rvert - c)$ otherwise	Huber loss: quadratic center, linear tails.
Robust inference	$\hat{\Sigma}_{ii} = e_i^2/(1-h_{ii})^2$ in HC3 meat	Heteroskedasticity-consistent covariance.
Transformation	$y^{(\lambda)} = (y^\lambda - 1)/\lambda$, with $\lambda = 0$ limit $\log y$	Box–Cox positive-response family.
Transformation	Yeo–Johnson piecewise power transform for $y \geq 0$ and $y < 0$	Power transformation allowing zero/negative values.
Likelihood/MLE	$L(\theta) = f(x \mid \theta)$, $\ell = \sum_i \log f(x_i \mid \theta)$, $\hat{\theta} = \arg\max \ell$	Fit a full probability model.
MLE uncertainty	$\mathrm{Var}(\hat{\theta}) \approx [-\ell''(\hat{\theta})]^{-1}$; vector case $[-\nabla^2\ell(\hat{\theta})]^{-1}$	Curvature/Hessian covariance approximation.
Delta Method	$\mathrm{Var}(g(\hat{\theta})) \approx G\Sigma G^T$	Propagate covariance through nonlinear derived quantities.
GARCH	$\sigma_t^2 = \omega + \sum_{i=1}^p \alpha_i X_{t-i}^2 + \sum_{j=1}^q \beta_j\sigma_{t-j}^2$	Dynamic conditional variance.

Function	Formula	Use
GARCH(1,1)	$\sigma_t^2 = \omega + \alpha X_{t-1}^2 + \beta\sigma_{t-1}^2$, stationary start $\omega/(1-\alpha-\beta)$	Core volatility recursion.
Nonstandard t	$\mathrm{Var}(X) = \sigma^2\nu/(\nu-2)$ for $X = \mu + \sigma T_\nu$	Distinguish scale parameter from SD.
VaR/CVaR	$\mathrm{VaR}_\alpha = \mu + \sigma z_\alpha$, $\mathrm{CVaR}_\alpha = \mu + \sigma\phi(z_\alpha)/\alpha$	Normal-model tail threshold and expected tail loss.

One final chain. Model assumptions determine the fitting rule; the fit produces estimators; their sampling behavior determines inference; diagnostics test whether the assumptions are credible; selection controls flexibility; repairs target specific failures; prediction and financial risk then use the fitted model while carrying forward the relevant uncertainty.

ABOUT THE AUTHOR

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I'm Mizuki Seika, a quantitative trader working at the intersection of mathematics, financial markets, and emerging technology; an explorer of Web3 and cryptocurrency; and, in a parallel universe that happens to be equally real, an independent idol and original music producer.

My academic path spans Carnegie Mellon University's Master of Science in Computational Finance (MSCF '28), Mathematics at Tsinghua University ('25), and an exchange year at the University of Oxford ('24). My interests range from quantitative research and systematic trading to decentralized finance and digital assets.

To me, markets and music have more in common than they appear to: both reward an ear for signals, a sense of structure, and the courage to build something original.

> *In finance, I turn complexity into models.*
> *In music, I turn it into melodies.*

This work is where I bring that same creative discipline to learning --- reconstructing knowledge until the pieces form a framework that actually makes sense.

Explore My Worlds

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Open-source projects and research

Spotify
Original music

Instagram
The creative side, on and off stage

Thank you for reading.